# Assessing the Relationship between Climate Variables and Rice Crop Yield in selected provinces in Northern Luzon Using Gaussian Process Regression

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**Abstract** Rice cultivation holds critical importance in the nation's agricultural landscape, addressing socio-cultural, economic, and political dimensions. With the Philippines identified as highly vulnerable to climate change, understanding the relationship between climate variables and rice crop yield becomes imperative for ensuring food security and sustainable production. Smallholder farmers, especially susceptible to climate changes, require effective mechanisms for adapting to evolving conditions. This study focuses on selected provinces in Northern Luzon, employing Gaussian Process Regression (GPR), a nonparametric machine learning model, to assess the relationship between climate variables and rice crop yield.

Data collection spans 2012-2020, integrating historical climate data and rice crop yield statistics. A collinearity matrix reveals complex relationships among climate variables. Notably, strong correlations between relative humidity (RH) and other variables pose challenges in isolating individual contributions. GPR models, utilizing various kernel functions, were developed and evaluated for multiple provinces. Squared Exponential GPR demonstrated superior performance, providing accurate predictions with low Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE). The study delves into the implications of climate variables on optimal planting windows. Analysis and visualization suggest Semester 1 as the optimal planting window, aligning with higher average temperatures during that period. The findings emphasize the significance of temperature conditions, particularly maximum temperatures, in influencing rice crop yield.

In conclusion, this research contributes to the sustainable management of agricultural resources, offering valuable insights for farmers, policymakers, and authorities. By incorporating machine learning techniques, specifically GPR, the study enhances our understanding of climate-crop yield relationships, providing a foundation for informed decision-making in the face of a changing climate.

**Keywords:** MATLAB, Gaussian Progress Regression, Yield Prediction, Climate

## 1 INTRODUCTION

### 1.1 Background of the Study

Rice cultivation spans approximately 4.8 million hectares of land in the Philippines (Statista Research Department, 2022). This highlights the integral role of rice in the nation's agricultural landscape. Even its cultural significance is ingrained as meals across the country are deemed incomplete without its presence on the table. For millions of Filipinos, it is a constant and essential food source. Thus, rice became an unparalleled socio-cultural, economic, and politically sensitive commodity. Even the economic stability of the country and other various facets are linked to the yields of rice production, making it imperative to ensure its reliable and sustainable supply (Mamiit, R., et. al., 2021).

The Philippines is considered as one of the most vulnerable countries to climate change. The impacts of climate variability have consequences on agriculture and food security. According to Mamun, M., et. al., (2023), understanding climatic variability is crucial for ensuring food security and mitigating climate risks. Smallholder farmers are especially vulnerable to a changing climate because they often lack the necessary access to information, support, resources, to respond and adapt effectively (Schenk, T., et. al., 2019). This vulnerability can lead to decreased agricultural productivity, threatening food security and the livelihoods of farmers. Given the role of rice in addressing food demand and ensuring food security in the Philippines, there exists a critical need to develop effective mechanisms for forecasting rice crop yields based on climate changes. Accurate prediction of crop production

from different climate variables is essential in effectively managing the food security and economic resilience of agricultural countries (Monterola, C., et. al., 2023). A reliable machine learning-based algorithm to quantify and predict future rice crop yields would not only contribute to the sustainable management of agricultural resources, but also address the broader socio-economic implications associated with food production in a certain region.

Models for predicting rice yield were created through the application of various technologies such as machine learning, deep learning, data mining, and the analysis of satellite and aerial imagery alongside statistical approaches using climate data. Different studies have utilized diverse methods involving climate data, including Artificial Neural Networks (ANN), Support Vector Machine Regression (SVMR), and Multiple Linear Regression (MLR). Notably, Gaussian Process Regression (GPR) has demonstrated superior performance in developing models for predicting rice yield compared to alternative statistical and machine learning techniques (Ekanayake,P., et. al., 2022). Thus, this study aims to assess the relationship between climate variables and rice crop yield in selected provinces in Northern Luzon, Philippines using Gaussian Process Regression Model.

The Gaussian Process Regression (GPR) Model is recognized as a nonparametric approach in machine learning that utilizes Gaussian processes for regression tasks. It is particularly effective in addressing nonlinear problems with limited sample sizes. A Gaussian process is a set of random variables characterized by a finite number of subsets possessing a joint Gaussian distribution. This model is an adapted form of linear regression, incorporating latent variables to elucidate the response. All the previously mentioned models based on machine learning techniques (Artificial Neural Networks, Support Vector Machine Regression, and GPR) were implemented using the MATLAB environment. It was also found out that Gaussian Process Regression has demonstrated notable accuracy in predicting yields based on climate and soil data (Wickramasinghe, L., et al.,2021).

**Climate and Agriculture**

The interconnection between agriculture and climate change is evident across various dimensions, as climate change stands as the primary driver of both biotic and abiotic stresses, leading to detrimental effects on the agriculture of a given region. The impacts of climate changes on the land and its agricultural practices are diverse, encompassing factors such as variations in annual rainfall, average temperature fluctuations, occurrences of heat waves, adjustments in the behavior of weeds, pests, or microbes, global alterations in atmospheric carbon dioxide or ozone levels, and fluctuations in sea level. The dynamic nature of global climate variations has significantly captured the attention of scientists due to the adverse impacts on global crop production, posing a threat to food security worldwide. According to some anticipated reports, agriculture is identified as the most vulnerable activity significantly affected by climate changes. (Raza et al., 2019)

As the world grapples with the challenges of global warming and climate change, the significance of weather forecasting becomes paramount, particularly in addressing its devastating impact on global food production. A compelling illustration is the agricultural struggles induced by climate change in Africa, marked by unpredictable precipitation patterns. Furthermore, a publication from the World Resources Institute indicates that global agricultural productivity may face a 17% reduction by 2050. This decline is attributed to the degradation of several fertile farmlands caused by wildfires, hurricanes, tornadoes, and persistent flooding, depleting the soil of its essential nutrients for sustainable agriculture. (Jiva, 2023)

Farmers depend on climate patterns and weather predictions in agriculture to make decisions on which crops to cultivate and when to plant them. The effectiveness of most farming practices relies heavily on favorable meteorological conditions. For instance, precision irrigation is contingent on the amount of rainfall a field receives, while soil solarization necessitates high temperatures, among other factors. Typically, farmers rely on short-range forecasts, which provide predictions for a period of 1 to 7 days. However, medium-range (up to a month), long-range (up to a year), and forecasts for

hazardous weather in agriculture can be valuable for planning field activities. Despite the possibility of abrupt severe weather events that can wreak havoc on crops and impact the viability of stored seeds, local meteorological forecasts assist farmers in preparing for such occurrences, thereby minimizing or even eliminating their detrimental effects. (Ivanchuk, 2023).

The anticipated consequences of climate change on significant crop yields have been extensively cataloged, drawing from a body of research spanning two decades. Predominantly, adverse effects outweigh positive ones on a global scale. Assessments of climate trends' impact on crop production reveal that wheat and maize yields have already experienced detrimental effects in numerous regions and on a global scale. (Food and Agriculture Organization of the United Nations, 2015)

In previous projections from the Agricultural Model Intercomparison and Improvement Project's Global Gridded Crop Model Intercomparison, based on the Coupled Model Intercomparison Project Phase 5, considerable climate impacts on major crops were identified, but with substantial associated uncertainties. This presents updated projections for the twenty-first century using ensembles of the latest-generation crop and climate models. The results indicate notably more pessimistic yield responses for maize, soybean, and rice compared to the original ensemble. Maize productivity by the end of the century shifted from +5% to −6% (SSP126) and from +1% to −24% (SSP585), attributed to warmer climate projections and improved crop model sensitivities. In contrast, wheat shows stronger gains (+9% shifted to +18%, SSP585), linked to higher CO2 concentrations and expanded gains in high-latitude areas. Notably, the emergence of climate impacts consistently occurs earlier in these new projections—before 2040 for several main producing regions. While future yield estimates remain uncertain, these findings suggest that major breadbasket regions will confront distinct anthropogenic climatic risks sooner than previously anticipated. (Müller et al., 2021)

Anticipations of future climatic conditions reveal a notable uptick in temperature alongside unpredictable and intensified rainfall, marked by variability in predicting extreme climate events. Projections for the mid-century period (2040–2069) indicate an expected increase of 2.8°C in maximum temperature and 2.2°C in minimum temperature across Asia. Addressing the unfavorable consequences of these climate change scenarios necessitates the optimization of climate-smart and resilient agricultural practices and technology to ensure sustainable productivity. (Ahmad et al., 2022)

**Machine Learning Methods in Weather Forecasting**

Modern farming, propelled by rapid technological advancements such as precision agriculture, provides a platform to empower contemporary farmers with the necessary skills and technical knowledge. This empowerment enables farmers to navigate their agricultural endeavors successfully by incorporating weather forecasting into their practices. The benefits of weather forecasting are akin to possessing superior intelligence or the ability to preempt unfavorable situations, especially in the face of adverse weather conditions. Whether addressing issues like indiscriminate crop wilting or other challenges encountered by smallholder farmers, accurate weather prediction offers an unparalleled advantage over the elements and perceived opposing forces of nature. (Jiva, 2023).

Weather forecasting typically follows a two-phased approach, incorporating satellite-derived images for weather data and predicting variables such as wind speed, precipitation, humidity, temperature, and other factors crucial in determining abnormal weather changes. Numerous models have been presented in the literature, with a predominant focus on utilizing time-series data. Time-series models not only aid in weather forecasting but also contribute to understanding relevant feedback in sectors like stock marketing and agriculture. Various models, including linear regression, time series vector models, artificial neural networks, and data mining techniques, have been employed for weather prediction. However, a significant portion of weather data exhibits nonlinear hidden patterns in a vast dimensional space, making it challenging to model using regression functions. While parametric models have been emphasized in the literature, they face

limitations due to the rigid nature of atmospheric pressure structures, hindering their effectiveness in modeling weather data. To address these limitations, it becomes imperative to explore Non-Parametric Regression Models, such as Gaussian Regression Models. These models prove effective in identifying nonlinear patterns and managing extensive dimensional data more flexibly, offering a promising avenue for improved weather modeling. (Krishna, 2015).

In recent years, the Gaussian Process Regression (GPR) algorithm has proven successful in applications within remote sensing and Earth sciences. Renowned for its commendable computational efficiency, stability, simplicity, and overall robustness compared to other statistical regression tools, GPR stands out. Notably, it demands a relatively small training dataset, allowing for the adoption of highly flexible kernel functions, and exhibits the capacity to deliver satisfactory predictions. An illustrative example is the application of a GPR model to predict daily Global Solar Radiation (GSR), where the outcomes demonstrated superior performance compared to conventional methods. (Ferkous et al., 2021).

### 1.2 Statement of the Problem

The primary objective of this research was to employ the Gaussian Process Regression Model to assess the relationship between climate variables and rice crop yield within selected provinces in Northern Luzon, Philippines. In pursuit of this goal, the study concentrated on addressing specific problems as outlined below:

**Is the Gaussian Process Regression Model appropriate to develop a machine learning based algorithm to quantify and predict future rice crop yields given climate parameters?**

This study aims to ascertain the appropriateness of the Gaussian Process Regression Model as a viable tool for constructing a machine learning algorithm capable of quantifying and predicting future rice crop yields based on various climate parameters. These parameters encompassed factors such as rainfall, maximum, minimum, and mean temperatures, wind speed, and relative humidity. The research sought to determine the model's efficacy by employing evaluation metrics to validate the outcomes derived from the dataset.

**Does the optimal planting window for each province can be determined using the predicted data from Gaussian Process Regression Model?**

This study aims to know whether the Gaussian Process Regression Model could yield insights into the identification of optimal planting windows for each province throughout the year. This part aimed to contribute valuable information for optimizing agricultural practices based on the predictive capabilities of the Gaussian Process Regression Model.

### 1.3 Objectives of the Study

The following are the main objectives of the study:

1. Apply the Gaussian process regression model to develop a machine learning-based algorithm to quantify and predict future rice crop yields given climate parameters (amount of rainfall, maximum, mean, and minimum temperatures, wind speed, and relative humidity).

2. Determine the optimal planting window for each province using the predicted data.

### 1.4 Scope and Limitations of the Study

This study focuses on specific provinces in Northern Luzon, Philippines including provinces in the Ilocos Region, Cordillera Administrative Region, Cagayan Region, and the northern portion of Central Luzon. The primary analytical tool employed is the Gaussian Process Regression Model, utilized to assess and predict rice crop yields. The research specifically addresses climate variables such as rainfall, maximum, minimum, and mean temperatures, wind speed, and relative humidity. These parameters will be analyzed to understand their impact on rice crop yield. On the other hand, there are also certain limitations that may impact the scope and generalizability of the findings. The accuracy and reliability of the study depend on the availability and quality of historical data on climate variables and rice crop yields obtained from the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA)

from 2012-2020. Limitations in data accessibility or gaps in historical records may affect the analysis. Moreover, this study focuses on selected provinces in Northern Luzon, and therefore, the findings are not directly extrapolated to other regions with distinct climatic and agricultural characteristics. Furthermore, the effectiveness of the Gaussian Process Regression Model, as a machine learning technique, is subject to its sensitivity to input parameters and the assumptions inherent in the model. Variability in model sensitivity may influence the accuracy of predictions.

### 1.5. Significance of the Study

This study holds significance for various stakeholders in the Philippine agricultural landscape, addressing crucial issues related to food security and sustainable rice production under the threat of climate change. By developing a machine learning model using Gaussian Process Regression (GPR), this study contributes to improved prediction of future rice crop yields based on climate parameters. This empowers farmers to make informed decisions regarding planting times, resource allocation, and risk management, potentially leading to increased productivity and income. Identifying optimal planting windows based on predicted climate patterns allows farmers to adjust their practices, minimize losses due to unpredictable weather events, and build resilience against the long-term impacts of climate change.

Moreover, reliable rice yield forecasts contribute to increased national food security. By providing valuable insights for policymakers and agricultural authorities, this study can inform strategies for food production planning, resource allocation, and import/export decisions, ultimately ensuring a stable supply of rice for the growing Filipino population. The GPR model and its predictive capabilities can promote sustainable agricultural practices. Efficient resource utilization and informed planting decisions based on climate forecasts can minimize environmental impact and contribute to long-term sustainability in the rice production sector.

Furthermore, this research contributes to the advancement of machine learning applications in agriculture. The successful implementation of GPR for rice yield prediction paves the way for further exploration of data-driven approaches to address diverse agricultural challenges, fostering innovation and technological progress in the field. By addressing these critical issues, this study holds significant potential to benefit farmers, policymakers, and the entire agricultural sector in the Philippines, promoting food security, climate resilience, and sustainable rice production in the face of a changing climate.

### 1.6. Definition of Terms

**Gaussian Process Regression (GPR) Model:** A nonparametric machine learning approach that uses Gaussian processes for regression tasks. It is effective in handling nonlinear problems with limited sample sizes.

**Optimal Planting Window:** The time frame during which conditions are most favorable for planting crops, considering climate variables such as rainfall, temperature, wind speed, and relative humidity.

**Kernel Function**: A mathematical function used in Gaussian Process Regression (GPR) to model the covariance between data points. In this study, four kernel functions were employed: Rational Quadratic, Exponential, Squared Exponential, and Matern 5/2.

**Production Volume:** The quantity of rice produced, measured in metric tons.

**Area Harvested:** The total land area from which rice is harvested, measured in hectares.

**BIAS:** A measure of the systematic error in the model predictions.

**MAPE (Mean Absolute Percentage Error):** A measure of the accuracy of the model, calculated as the average percentage difference between predicted and actual yields.

**MSE (Mean Squared Error):** A measure of the average squared difference between predicted and actual yields.

**RSR (Root Mean Squared Error Ratio):** The ratio of the root mean squared error to the standard deviation of the observed yields.

**Nash Number:** A statistic used to assess the performance of the models.

**RRMSE (Relative Root Mean Square Error):** A normalized measure of the root mean squared error.

**Climate Variables:** Factors such as rainfall, temperature, wind speed, and humidity that influence the growth and development of crops.

**Machine Learning:** A subfield of Artificial Intelligence that allows computers to learn from data without being explicitly programmed, used for tasks like prediction and classification.

### 1.7. Research Framework

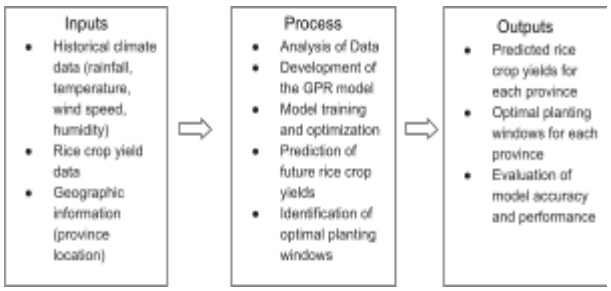



*Figure 1. Schematic diagram of the Research Framework.*

## 2 METHODOLOGY

### 2.1 Area of the Study

This study is focused on the rice growing provinces of Northern part of Luzon (Figure 2). This area includes provinces in the Ilocos Region (Ilocos Norte, Ilocos Sur, La Union and Pangasinan), Cordillera Administrative Region (Abra, Apayao, Benguet, Ifugao, Kalinga, Mountain Province), Cagayan region (Batanes, Cagayan, Isabela, Nueva Vizcaya, Quirino), and the northern portion of Central Luzon, specifically Aurora, Nueva Ecija, Tarlac, and Zambales.

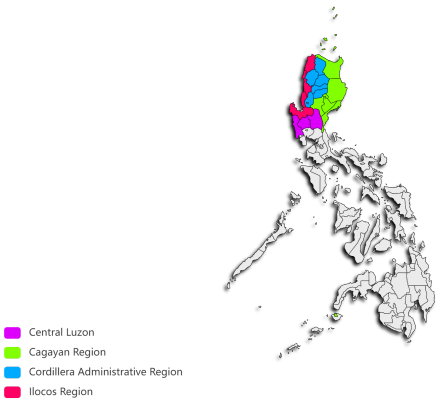



*Figure 2. Map of studied area. Map of the agricultural area of Northern Luzon*

### 2.2 Data Collection

#### 2.2.1 Climatological Data

Historical climate data from 2012-2020 are gathered from Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) for the provinces in Northern Luzon (PAGASA, n.d.). For provinces with no recorded data in PAGASA, we acquired monthly climate data from 2012-2020 from the website, Weather and Climate - The Global Historical Weather and Climate Data (Philippines Climate Zone, n.d.).

Based on the distribution of rainfall, Northern Luzon has four distinct climatic types: Type I, which has two distinct seasons—the dry season from November to April and the wet season the rest of the year; Type II, which has no dry period but the highest rainfall from November to January; and Type III, which has two distinct seasons—Dry from November to February and wet the rest of the year (Gutierrez et al., 2019).

The climatic factors/variables taken into account for modeling in the climatological data are wind speed, relative humidity, rainfall amount, maximum temperature, and minimum temperature. The average of this climate data was calculated for every season in every province. The following equation represents the nonlinear relationship between the climatological parameters and rice yield:

rice yield=(RF, Tmin, Tmax, T,RH, AWS) ,

where RF is the rainfall amount in mm, Tmin is the minimum temperature in °C, Tmax is the maximum

temperature in °C, RH is the relative humidity in %, and AWS is the average wind speed in mps.

| Group | Variable | Notation | Unit |
|---|---|---|---|
| Agriculture | Rice Yield | RY | metric tons/hect |
| Climatological Data | Rainfall Amount | $RAINFALL$ | mm |
| | Minimum Temperature | $T_{min}$ | °C |
| | Maximum Temperature | $T_{max}$ | °C |
| | Mean Temperature | $T_{mean}$ | °C |
| | Relative Humidity | $RH$ | % |
| | Wind Speed | $WS$ | mps |
| Management practices | Planting seasons (Semester 1: Dry Season, Semester 2: Wet Season) | | Semester |

Table 1. Summary of Data used in the study

**2.2.2 Rice Crop Yield Data**

Rice crop yield data is not readily available in the dataset from Philippine Statistics Authority (PSA). It was calculated using the following formula:

$$Rice\ Crop\ Yield = \frac{Production\ Volume\ (metric\ tons)}{Area\ Harvested\ (hectares)}$$

Datasets for each semester from 2012-2020 for the rice production volume of both irrigated and rainfed, and the area harvested of the provinces in Northern Luzon are obtained from PSA Openstat Website (PSA, n.d. ).

Two main agricultural seasons were considered, namely, the dry and wet season. The planting window for Semester 1, which is the dry season, is from September to February, and for Semester 2, the wet season, is from April to August (Gutierrez et al., 2019).

**2.3. Machine Learning Model Development**

This section constitutes the methods used in the preprocessing and analysis of data, the machine learning method algorithm used in the study—Gaussian Process Regression, and the evaluation metrics used to validate the result of the data.

**2.3.1 Analysis of Data**

Pairwise correlations are utilized to identify the influence of the climate index on the yield. The correlation between each independent variable was inspected to identify their collinearity. The Pearson correlation coefficient was computed using MATLAB, the programming language used in this study.

Coefficients tending toward 1, suggests strong interdependence among the variables. Values close to 0 imply a lack of a distinct relationship. In this study, a robust correlation between two indices were defined if the correlation value was within the interval [0.75,1.0]. Moderate correlations were identified when one value ranged from 0.50 to 0.74 while the other falls in the higher range.

**2.3.2. GPR Modeling for Predicting Rice Crop Yield**

The Gaussian Process Regressions, or GPRs, address a limited quantity of random variables with a distribution that is multivariate. Gaussian processes are influenced by the Normal distribution concept. As a non-parametric machine learning technique, the GPR performs Gaussian processes for tasks linked to regression and is particularly useful in handling non-linear issues with small sample sizes. It makes use of latent variables to project the answer. The Gaussian process that will be applied on a training data set, which is defined as $\{(x_g, y_g): g = 1, 2,..., n\}$ where, $x_g \in R^d$ and $y_g \in R$, is given as follows:

$$y \sim GP(m(x), k(x_g, x_h))$$

where $m(x)$ is the mean function denoting the expectation and $k(x_g, y_g)$ is the Kernel function. The symmetrical Kernel function represents the covariance. This is a collection of random variables that share features with a joint Gaussian distribution. The covariance matrix of a linear regression model in the form of $y = x^T\beta + \varepsilon$ is parametrized as follows:

$$\begin{pmatrix} k(x_1,x_1) & k(x_1,x_2) & \cdots & k(x_1,x_n) \\ k(x_2,x_1) & k(x_2,x_2) & \cdots & k(x_2,x_n) \\ \vdots & \vdots & \vdots & \vdots \\ k(x_n,x_1) & k(x_n,x_2) & \cdots & k(x_n,x_n) \end{pmatrix}$$

Utilizing the climate data, prediction models for rice yield were developed in this project. Four kernel functions—Rational Quadratic, Exponential, Squared Exponential, and Matern 5/2—were employed with the GPR, which was modeled in MATLAB.

Rational Quadratic GPR Kernel is used for modeling data that varies at multiple scales and is appropriate for multivariate statistical analysis. The Rational Quadratic Kernel function is given by the following equation:

$$k(x_g, x_h|\theta) = \sigma_f^2(1 + \frac{r^2}{2\alpha\sigma_1^2})$$

$$k(x_g, x_h|\theta) = \sigma_f^2(1 + \frac{r^2}{2\alpha\sigma_1^2})$$

where,

$$r = \sqrt{(x_g - x_h)^T(x_g - x_h)},$$

$\theta$ is the maximum a posteriori estimates, $\sigma_f$ is the standard deviation, $\sigma_1$ is the length scale, $T$ is the transpose, and $\alpha$ is the non negative parameter of the covariance.

Squared Exponential GPR is a function of a regression model with an infinite number of RBFs. The Square Exponential Kernel function is written as follows

$$k(x_g, x_h|\theta) = \sigma_f^2 exp(-\frac{r^2}{2\sigma_1^2}).$$

Matern 5/2 Kernel takes spectral densities of the stationary Kernel and performs Fourier transformation of the RBF Kernel. Its Kernel function is expressed as

$$k(x_g, x_h|\theta) = \sigma_f^2(1 + \frac{\sqrt{3r}}{\sigma_1})exp(-\frac{\sqrt{3r}}{\sigma_1}).$$

Exponential GPR manages smooth function well with minimal errors though not efficient in handling discontinuities. The Kernel function of the Exponential GPR is given as

$$k(x_g, x_h|\theta) = \sigma_f^2 exp(-\frac{\sqrt{3r}}{\sigma_1}).$$

### 2.3.3 Evaluation Metrics

The GPR-based crop-climate models were evaluated using the following statistical parameters:

BIAS:

$$BIAS = \frac{\sum_{i=1}^{N}(P_i - A_i)}{N}$$

Mean Absolute Percentage Error:

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}(\frac{A_i - P_i}{A_i})100\%$$

Mean Squared Error:

$$MSE = \frac{\sum_{i=1}^{N}(A_i - P_i)^2}{N}$$

Root Mean Squared Error Ratio:

$$RSR = \frac{\sqrt{MSE}}{\sigma_A}$$

Nash Number:

$$Nash\ Number = 1 - [\frac{\sum_{i=1}^{N}(A_i - P_i)^2}{\sum_{i=1}^{N}(A_i - \overline{A}_i)^2}]$$

Relative Root Mean Square Error:

$$RRMSE = \sqrt{\frac{1}{N}\sum(\frac{A_i - P_i}{A_i})^2}$$

where, $A$ is the actual yield and $P$ is the predicted yield, $\overline{A}$ and $\overline{P}$ are the means, $\sigma_A$ is the standard deviation, and $N$ is the total number of data points. The most suitable models for yield estimation are indicated by low values of MSE, MAPE, RSR, and RRMSE, according to statistical norms. In the same way, highly accurate models have a Pearson's correlation coefficient and a Nash number that are closer to 1 (Ekanayake et al., 2022).

### 2.4 Predicting Optimal Planting Window

Developing the MATLAB code to predict an optimal planting window for crops involves several key steps, including data acquisition, preprocessing, analysis, and visualization. First was the acquisition of relevant data. A CSV file was used, containing columns representing key climate variables (rainfall, maximum temperature, minimum temperature, mean temperature, relative humidity, wind speed) and crop yield over nine years.

Upon obtaining the dataset, the data preprocessing phase involved reading the CSV file using MATLAB's readtable function. The code extracted individual columns for year, month, climate variables, and crop yield. Subsequently, the climate variables were combined into a matrix, facilitating further analysis.

Seasonal segmentation was performed based on months, creating separate indices for Semester 1 (January to June) and Semester 2 (July to December). The analysis phase focused on computing the average values of climate variables for each semester using the mean function in MATLAB. This involved calculating mean values for rainfall, maximum temperature, minimum temperature, mean temperature, relative humidity, and wind speed for both Semester 1 and Semester 2. Additionally, the average crop yield for each semester was determined. These computations served to provide insights into the seasonal trends in climate variables and their potential impact on crop yield. To visually represent the findings, two distinct figures were generated using MATLAB's plotting functionalities.

The first figure presented a subplot displaying the average climate variables for each semester, providing a comparative view of the seasonal variations. The second figure showcased a 3D scatter plot illustrating the relationship between specific climate variables (rainfall, TMAX) and crop yield. This visualization aimed to identify patterns and potential correlations between climate conditions and crop productivity. The final step involved determining the optimal planting semester based on the calculated average crop yields for Semester 1 and Semester 2.

## 3 RESULTS AND DISCUSSION

### 3.1 Collinearity of Weather Variables and Crop Yields

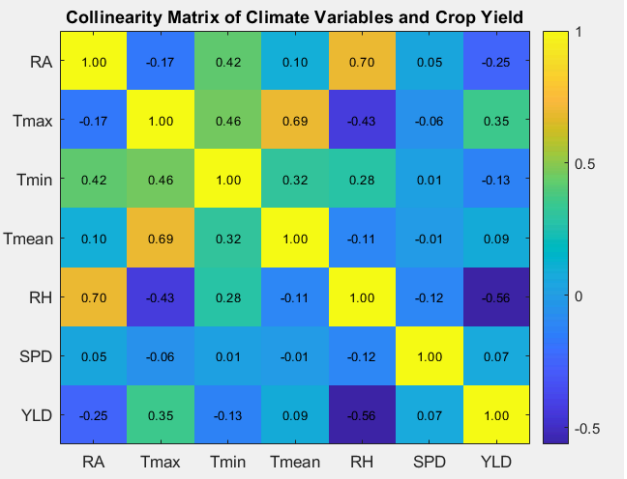



*Figure 3. Correlation Matrix of Weather Variables and Crop Yield of Abra*

The collinearity matrix of climate variables and crop yield presented in Figure 3, reveals intricate relationships among the weather parameters. These correlations, quantified through Pearson's correlation coefficients, provide valuable insights into the potential collinearity between climate factors and their impact on crop yield. The values in the matrix range from -0.5 to 1.00, representing the strength and direction of the linear relationships between the variables.

The collinearity matrix presented in Table 1 explores the interrelationships among various climate variables (RA, Tmax, Tmin, Tmean, RH, SPD) and crop yield (RY). Each cell in the matrix contains the Pearson correlation coefficient between the

respective pairs of variables. The correlation coefficient between RA and Tmax is -0.17, indicating a weak negative correlation. RA shows a moderate positive correlation with Tmin (0.42) and RH (0.70) and a weak positive correlation with Tmean (0.10). The negative correlation with SPD (-0.25) suggests a slight inverse relationship. Tmax exhibits a weak negative correlation with RA (-0.17) and a moderate positive correlation with Tmin (0.46) and RH (0.43). The negative correlation with SPD (-0.06) suggests a mild inverse relationship. Tmin shows a moderate positive correlation with RA (0.42) and Tmax (0.46) and a weak positive correlation with RH (0.28).

The negative correlation with SPD (-0.13) suggests a slight inverse relationship. Tmean displays a weak positive correlation with RA (0.10) and Tmax (0.69) and a slight negative correlation with RH (-0.11). The correlation with other variables is relatively weak. RH exhibits a strong positive correlation with RA (0.70) and Tmax (0.43), and a weak positive correlation with Tmin (0.28) and Tmean (0.11). The negative correlation with SPD (-0.56) is indicative of a substantial inverse relationship. SPD shows weak correlations with most climate variables, indicating limited associations with RA (0.05), Tmax (-0.06), Tmin (0.01), Tmean (-0.01), and RY (0.07). Crop yield (RY) has weak negative correlations with RA (-0.25), Tmax (0.35), Tmin (-0.13), Tmean (0.09), and SPD (0.07). These correlations suggest that crop yield may be influenced by a combination of climate factors, but the relationships are not excessively strong.

The collinearity matrix reveals several noteworthy associations among the climate variables and crop yield. Of particular interest is the strong positive correlation between rainfall (RA) and relative humidity (RH), suggesting a potential interdependence between these variables. Additionally, the negative correlations between crop yield (RY) and both RA and RH indicate that higher rainfall and humidity may have adverse effects on crop yield. The collinearity matrix provides insights into potential multicollinearity issues among climate variables. Notably, strong correlations between RH and RA, Tmax, and SPD, as well as the inverse correlation between RH and SPD, could pose challenges in isolating the individual contributions of these variables in predicting crop yield. Addressing multicollinearity may be essential to enhance the robustness of the regression model. Techniques such as variable selection, data transformation, or the application of ridge regression may be considered to mitigate these issues (Kutner et al., 2005). This highlights the importance of carefully examining collinearity in climate-crop yield relationships, ensuring the reliability and accuracy of the conclusions drawn from regression analyses.

### 3.2 Yield-Weather GPR Models

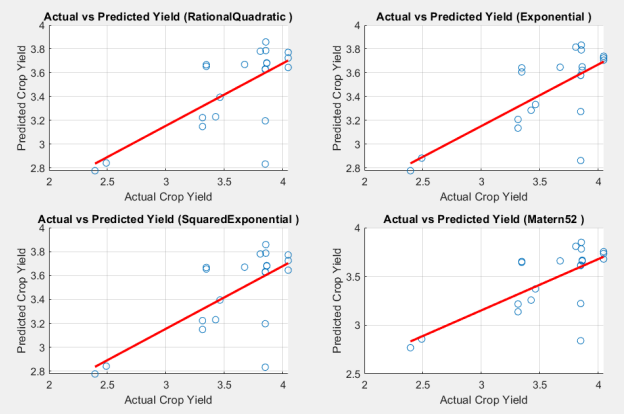



*Figure 4. Comparison of the Observed Rice Crop Yield in Abra in various GPR models (upper left: Rational Quadratic, upper right Exponential,lower left: Squared Exponential, lower right: Matern 5/2).*

Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | 0.062567 | 3.0963 | 0.027389 | 0.1655 | 0.44063 | 0.72988 |
| {'Exponential' } | 0.063137 | 3.1291 | 0.028547 | 0.16896 | 0.41699 | 0.74515 |
| {'SquaredExponential'} | 0.061477 | 3.0746 | 0.025902 | 0.16094 | 0.47099 | 0.7098 |
| {'Matern52' } | 0.062531 | 3.0899 | 0.027468 | 0.16573 | 0.43902 | 0.73094 |

*Figure 5. Evaluation Metrics for GPR Models computed by MATLAB.*

For all the provinces considered in this study, four GPR models were established based on the four Kernel functions: Rational Quadratic GPR, Exponential GPR, Squared Exponential GPR, and Matern 5/2 GPR. Evaluation metrics were used to determine the effectivity of the correlation performed by each Kernel function. The pioneer province used is Abra, and the performance of each model is illustrated by Figure 3. The evaluation metrics showed that Squared Exponential GPR outperformed the other three Kernel models, which yielded Mean Squared Error (MSE) = 0.0259, and

Mean Absolute Percentage Error (MAPE) of 3.0746. The other GPR models were also deemed significant presenting evaluation metrics values close to 0—Rational Quadratic GPR: MSE = 0.0274, MAPE = 3.0963, Exponential GPR: MSE = 0.0285, MAPE = 3.1291, Matern 5/2 GPR: MSE = 0.0275, MAPE = 3.0899.

The computed evaluation metrics for each GPR model established its accuracy and efficiency in predicting climate data. The GPR program is pioneered in Abra. Yield-weather GPR Models are also constructed in all the provinces from Northern Luzon (see Appendix A).

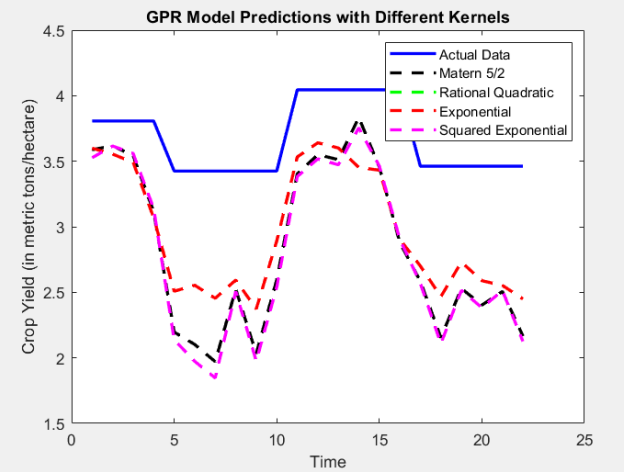



*Figure 6. Rice Crop Yield in Abra estimated by the four GPR models.*

Furthermore, Figure 6 visually represents the estimated rice crop yield in Abra using the four GPR models. The visual comparison shows the differences and similarities among the models, supporting the quantitative findings presented in Table 2. The results demonstrate that all four GPR models exhibit a commendable ability to predict the trend of the actual rice crop yield data in Abra. Each model demonstrates a meaningful correlation, as evidenced by their low Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) values. This indicates a strong predictive capability for each model, emphasizing that they all follow the observed trend of rice crop yield in Abra reasonably well. While all four GPR models demonstrate effectiveness in capturing the trend of the actual data, the Squared Exponential GPR model exhibits a slightly superior performance, suggesting its potential as a favorable choice for predicting rice crop yield.

### 3.3 Predicted Optimal Planting Window

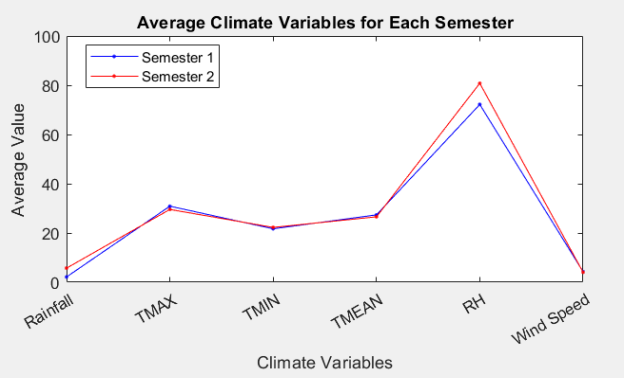



*Figure 7. Average Climate Variables for Each Semester in Abra.*

*See Appendix B for the Optimal Planting Window of other Provinces.

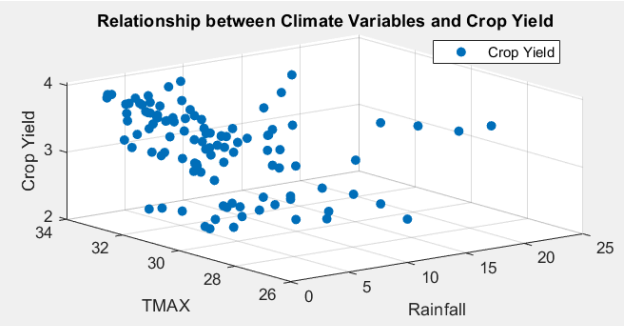



*Figure 8. Average Climate Variables for Each Semester in Abra.*

```
>> windows
The Optimal Planting Semester is Semester 1
```

*Figure 9. Optimal Planting Semester as Predicted by MATLAB.*

The results obtained from the analysis and visualization of climate variables in relation to crop yield provide valuable insights into the optimal planting semester for crops. The findings are summarized based on the observations derived from Figure 1 and Figure 2.

Figure 1 illustrates the average values of key climate variables for Semester 1 and Semester 2. Notably, Semester 1 exhibited higher average values for maximum temperature (TMAX) and mean temperature (TMEAN). In contrast, Semester 2 displayed higher average values for rainfall,

minimum temperature (TMIN), and relative humidity (RH). Wind Speed values (SPD) remained consistent across both semesters. This divergence in climate variables suggests distinct seasonal characteristics between Semester 1 and Semester 2.

Figure 2, a 3D scatter plot, depicts the relationship between specific climate variables (rainfall and maximum temperature) and crop yield. The plot reveals that higher crop yields are associated with elevated maximum temperatures (TMAX) rather than increased rainfall. This suggests that, within the context of the provided dataset, crop yield is more influenced by temperature conditions, specifically higher maximum temperatures.

Based on the analysis, the code concludes that Semester 1 is the optimal planting window for crops (Figure n). This conclusion is supported by the higher average values of critical climate variables such as maximum temperature and mean temperature during Semester 1. The findings align with the observed trend in the 3D scatter plot, where maximum temperature (TMAX) appears to have a more significant positive correlation with crop yield compared to rainfall.

## 4 SUMMARY, CONCLUSION, AND RECOMMENDATIONS

### 4.1 Summary

This study, entitled "Assessing the Relationship between Climate Variables and Rice Crop Yield in Selected Provinces in Northern Luzon Using Gaussian Process Regression," aimed to apply the Gaussian process regression model to develop a machine learning-based algorithm to predict future rice crop yields given several climate parameters such as the amount of rainfall, temperature, wind speed, and relative humidity. Likewise, it sought to determine the optimal planting window for each province using the predicted data.

The first objective was addressed by determining the collinearity among the variables. This study used the Pearson correlation coefficient. Using four different Gaussian Process Regression (GPR) models: Rational Quadratic GPR, Squared Exponential GPR, Matern 5/2, and Exponential GPR, the most appropriate model for each province was used to predict the future yield of rice crops given climatological parameters. Moreover, the study was able to assess these machine learning-based models through evaluation metrics - BIAS, Mean Absolute Percentage Error, Mean Squared Error, Root Mean Squared Error Ratio, Nash Number, and Relative Root Mean Square Error.

This study was able to predict the optimal planting window through MATLAB. The key steps include data acquisition, preprocessing, analysis, and visualization. Figures were also utilized to visually represent the findings. Through this, the study identified patterns and potential correlations between climate conditions and crop productivity and likewise determined the optimal planting semester.

### 4.2 Conclusion

In conclusion, this research focused on assessing the relationship between climate variables and rice crop yield in selected provinces in Northern Luzon, Philippines, utilizing the Gaussian Process Regression (GPR) Model. The study addressed key questions regarding the appropriateness of the GPR model for developing a machine learning algorithm to predict future rice crop yields and determining optimal planting windows for each province. Four GPR models based on different kernel functions were developed and evaluated for their effectiveness in predicting rice crop yields. The results demonstrated the overall effectiveness of GPR models, with the Squared Exponential GPR model exhibiting slightly superior performance.

Moreover, the study explored the relationship between climate variables and crop yield, emphasizing the significance of temperature conditions, particularly maximum temperature, in influencing crop productivity. The analysis suggested that Semester 1, characterized by higher maximum temperatures, is the optimal planting window for crops based on the dataset. This study underscores the importance of leveraging machine learning models, particularly GPR, in enhancing the understanding of climate-crop yield relationships, ultimately contributing to the development of resilient and sustainable agricultural practices.

It is crucial to note that these conclusions are drawn from the analysis of the specific dataset used in this research. Further studies and validation with additional datasets and, if possible, field experiments are recommended to generalize these findings and refine the understanding of optimal planting windows for different crops. Additionally, the results emphasize the significance of considering multiple climate variables in crop planning and highlight the potential for utilizing such analyses in precision agriculture to enhance agricultural sustainability and productivity.

### 4.3 Recommendation

This comprehensive study on "Assessing the Relationship between Climate Variables and Rice Crop Yield in Selected Provinces in Northern Luzon Using Gaussian Process Regression" has provided valuable insights into the dynamics between climate variables and rice crop yield. The collinearity analysis, GPR modeling, and assessment of the optimal planting window offer a foundation for future research endeavors. In light of the findings, the following research recommendations are proposed:

1. The prospect of using other predictive machine learning techniques such as artificial neural networks and random forests is recommended by the researchers. Exploring and comparing the results of other techniques would give more accurate results.
2. Integrate additional crop related variables, including soil characteristics, crop management practices, or pest prevalence, into the analysis. Also, including other weather and climate data variables such as solar radiation, and climate extremes. Considering these factors could contribute to a more holistic understanding of the determinants of rice crop yield and facilitate the development of more accurate predictive models.
3. Expand the geographical scope of the research to have a broader range of provinces in the Philippines. Investigating climate-crop relationships in the different climate types in the Philippines would reveal specific nuances which can enable the development of strategies for improved crop yield in the Philippines.
4. Lastly, since the researchers only used nine (9) years of historical data in this study, increasing historical data of the provinces' climate and crop data would provide more accurate results in predicting climate and crop yields trend.

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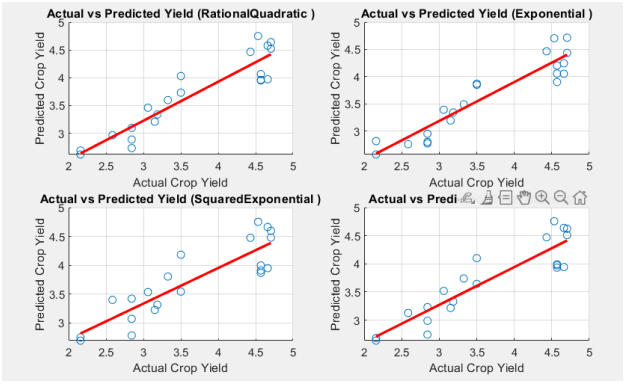
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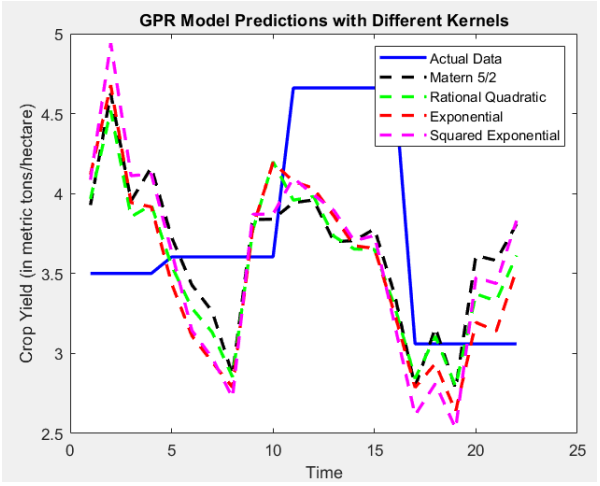
# 6 APPENDIX

## 6.1. Yield-Weather GPR Models for Each Province

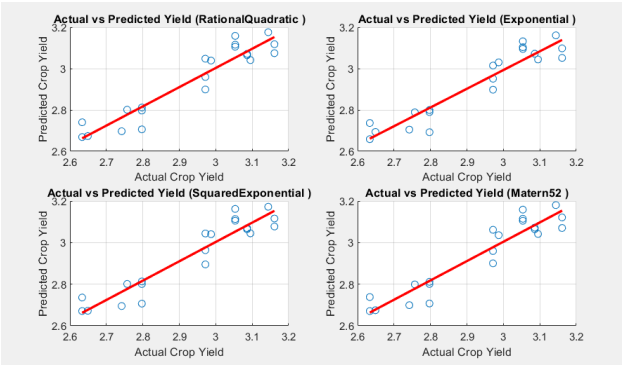
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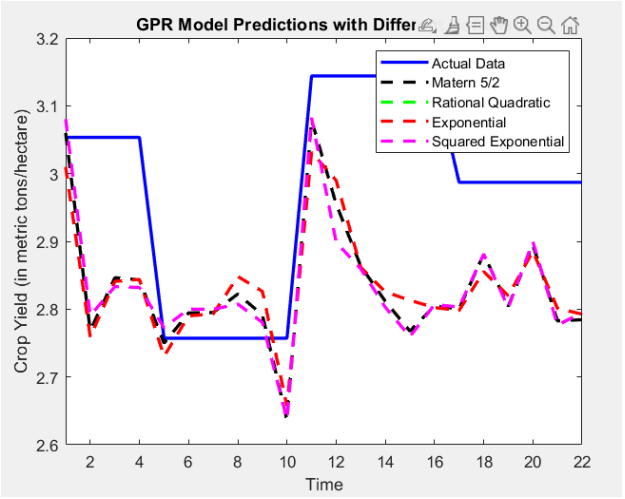

```
>> GPRmod
Evaluation Metrics for GPR Models:
        Kernel          BIAS      MAPE     MSE      RMSE     Nash     RRMSE
    {'RationalQuadratic' }    0.033304    9.0188    0.13703    0.37018    0.82745    0.40538
    {'Exponential'       }    0.0019182    8.2368    0.12203    0.34933    0.84634    0.38254
    {'SquaredExponential'}    0.089567    11.525    0.21483     0.4635    0.72949    0.50757
    {'Matern52'          }    0.059775    10.096    0.16525    0.40651    0.79193    0.44516
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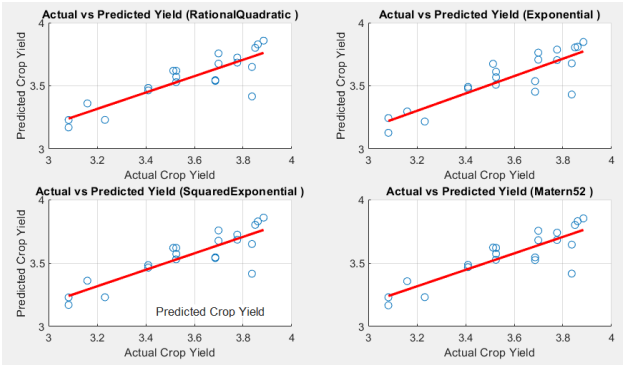

### Benguet




```
Evaluation Metrics for GPR Models:
        Kernel          BIAS      MAPE     MSE      RMSE     Nash     RRMSE
    {'RationalQuadratic' }    0.0076642    1.6993    0.0033864    0.058192    0.89145    0.32153
    {'Exponential'       }    -0.00084239    1.5876    0.0031393    0.05603    0.89937    0.30958
    {'SquaredExponential'}    0.0076743    1.6916    0.0033431    0.05782    0.89284    0.31947
    {'Matern52'          }    0.0087551    1.7163    0.0034634    0.058851    0.88898    0.32517
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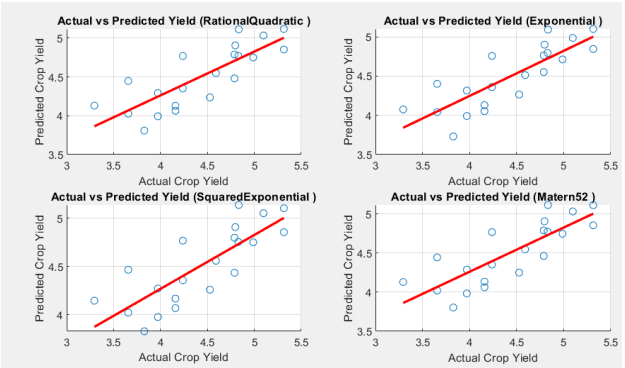



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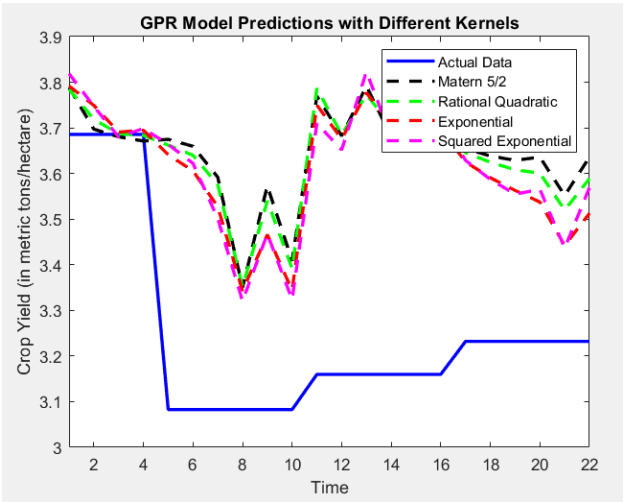


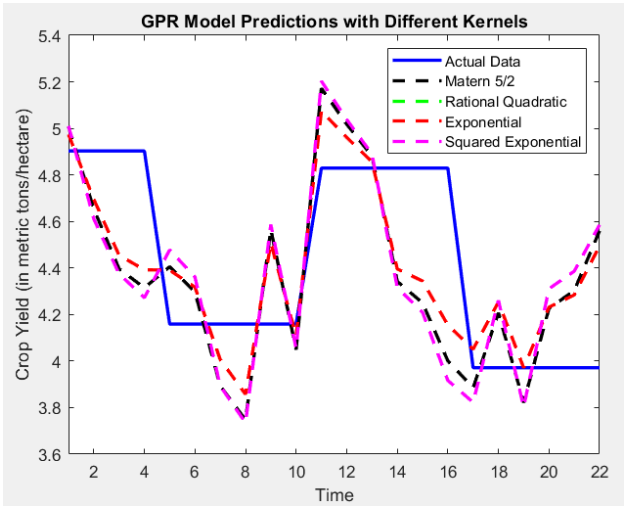

```
Evaluation Metrics for GPR Models:
        Kernel          BIAS      MAPE     MSE      RMSE     Nash     RRMSE
    {'RationalQuadratic' }    -0.014452    2.7292    0.017744    0.13321    0.72721    0.5097
    {'Exponential'       }    -0.015028    2.7303     0.01789    0.13375    0.72497    0.5118
    {'SquaredExponential'}    -0.014452    2.7292    0.017744    0.13321    0.72721    0.5097
    {'Matern52'          }    -0.014869    2.7327    0.017953    0.13399      0.724    0.51269
```

### Kalinga

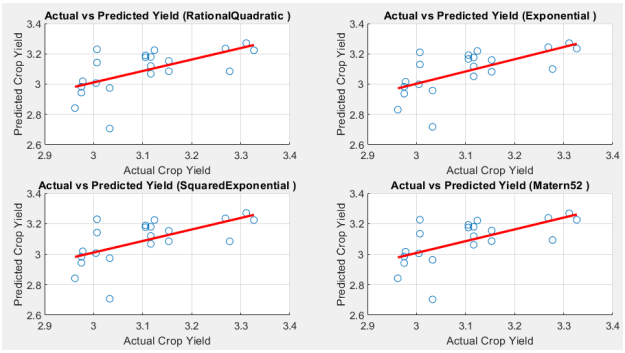



```
Evaluation Metrics for GPR Models:
        Kernel          BIAS      MAPE     MSE      RMSE     Nash     RRMSE
    {'RationalQuadratic' }    0.074965    5.9959    0.11702    0.34209    0.62548    0.59723
    {'Exponential'       }    0.064467    6.0154    0.10962    0.33109    0.64916    0.57804
    {'SquaredExponential'}    0.078832    5.999    0.12043    0.34702    0.61459    0.60585
    {'Matern52'          }    0.073623    5.98    0.11666    0.34156    0.62663    0.59631
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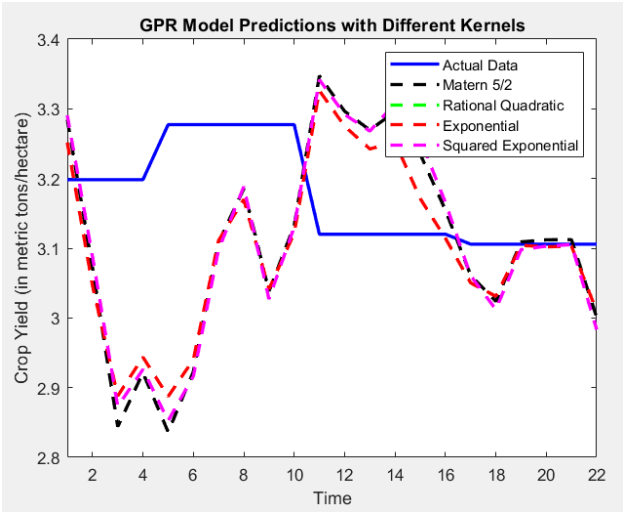




## Mountain Province

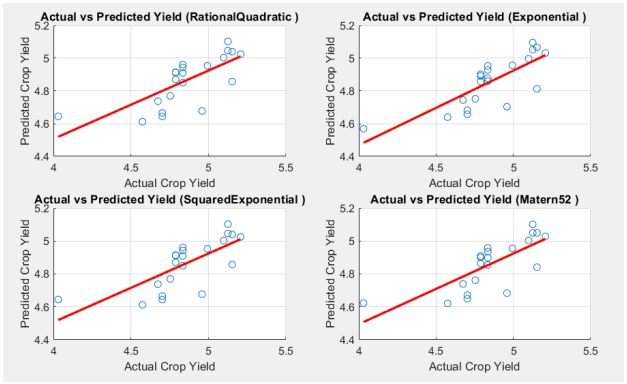



Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | -0.013994 | 2.7011 | 0.013262 | 0.11516 | -0.056284 | 1.003 |
| {'Exponential' } | -0.017218 | 2.6437 | 0.012242 | 0.11064 | 0.02495 | 0.96365 |
| {'SquaredExponential'} | -0.013994 | 2.7011 | 0.013262 | 0.11516 | -0.056284 | 1.003 |
| {'Matern52' } | -0.015157 | 2.6926 | 0.013105 | 0.11448 | -0.043763 | 0.99703 |

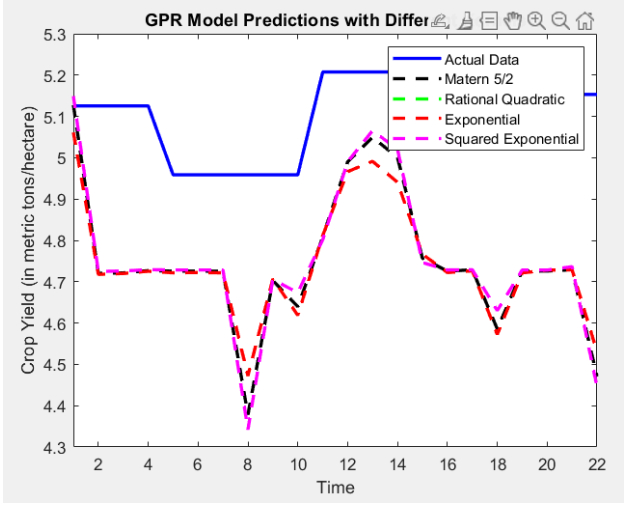



## Ilocos Norte

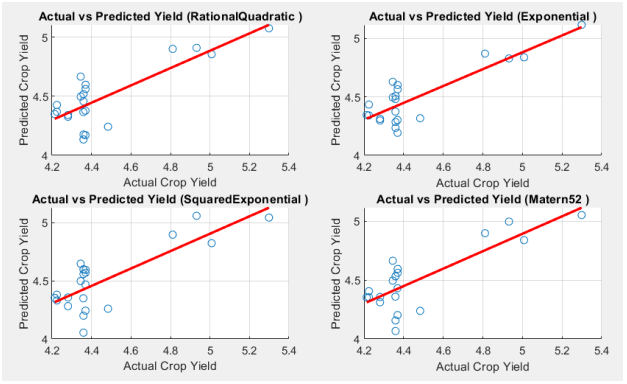



Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | 0.0086457 | 2.6275 | 0.032812 | 0.18114 | 0.50787 | 0.68461 |
| {'Exponential' } | 0.0031925 | 2.4274 | 0.028195 | 0.16791 | 0.57712 | 0.63462 |
| {'SquaredExponential'} | 0.0086457 | 2.6275 | 0.032812 | 0.18114 | 0.50787 | 0.68461 |
| {'Matern52' } | 0.0066259 | 2.5467 | 0.031272 | 0.17684 | 0.53097 | 0.66835 |

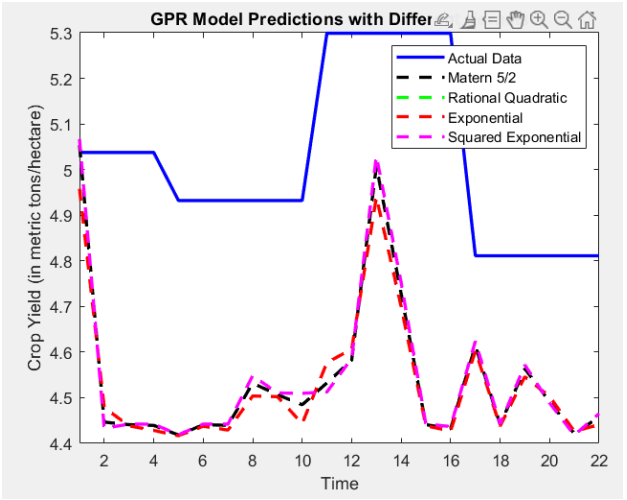


## Ilocos Sur

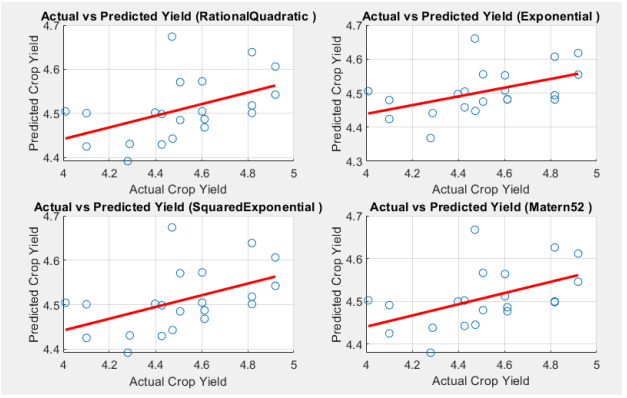



Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | 0.029233 | 3.3181 | 0.028607 | 0.16914 | 0.64585 | 0.58077 |
| {'Exponential' } | 0.030717 | 2.9648 | 0.022324 | 0.14941 | 0.72363 | 0.51304 |
| {'SquaredExponential'} | 0.03999 | 3.5863 | 0.032305 | 0.17974 | 0.60006 | 0.61716 |
| {'Matern52' } | 0.032902 | 3.5572 | 0.03176 | 0.17821 | 0.60682 | 0.61193 |

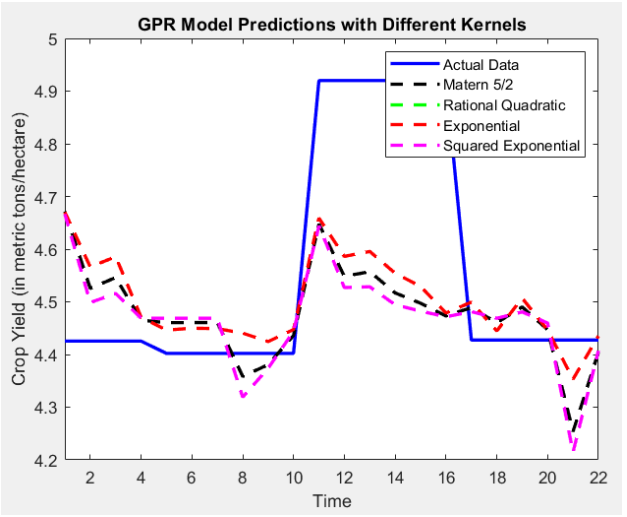



## La Union

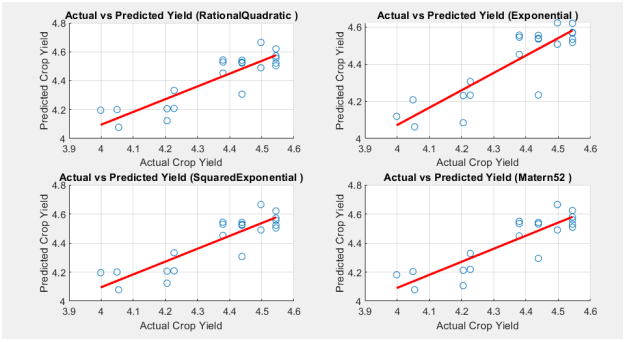



Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | -0.0010484 | 4.1105 | 0.053166 | 0.23058 | 0.18864 | 0.87904 |
| {'Exponential' } | -0.005938 | 4.1393 | 0.053147 | 0.23054 | 0.18893 | 0.87889 |
| {'SquaredExponential'} | -0.0010484 | 4.1105 | 0.053166 | 0.23058 | 0.18864 | 0.87904 |
| {'Matern52' } | -0.0027313 | 4.1169 | 0.052983 | 0.23018 | 0.19142 | 0.87754 |



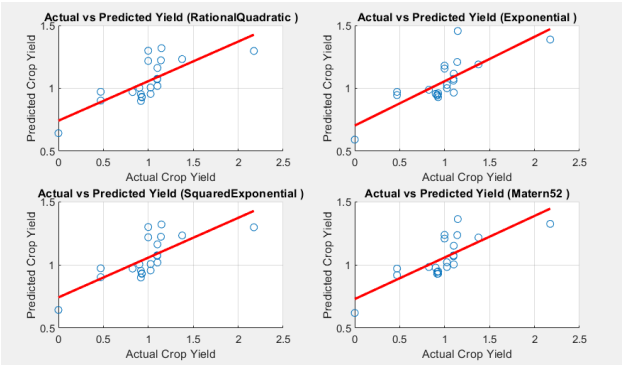

## Pangasinan




Evaluation Metrics for GPR Models:

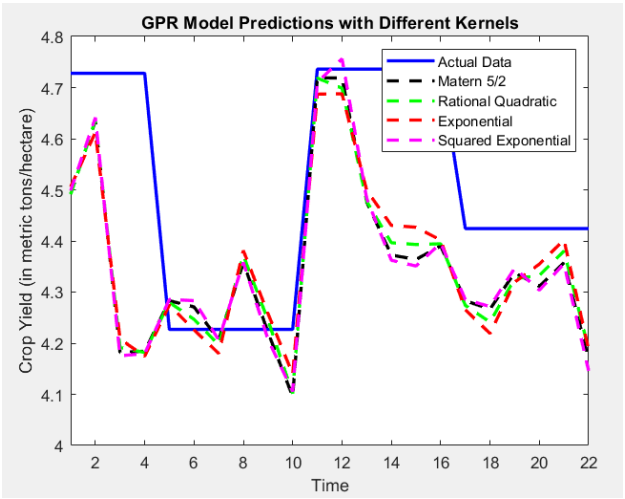
| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | 0.053525 | 1.9076 | 0.010261 | 0.1013 | 0.66123 | 0.56801 |
| {'Exponential' } | 0.049229 | 1.9285 | 0.010703 | 0.10346 | 0.64664 | 0.58011 |
| {'SquaredExponential'} | 0.053525 | 1.9076 | 0.010261 | 0.1013 | 0.66123 | 0.56801 |
| {'Matern52' } | 0.054397 | 1.9379 | 0.010558 | 0.10275 | 0.65146 | 0.57615 |

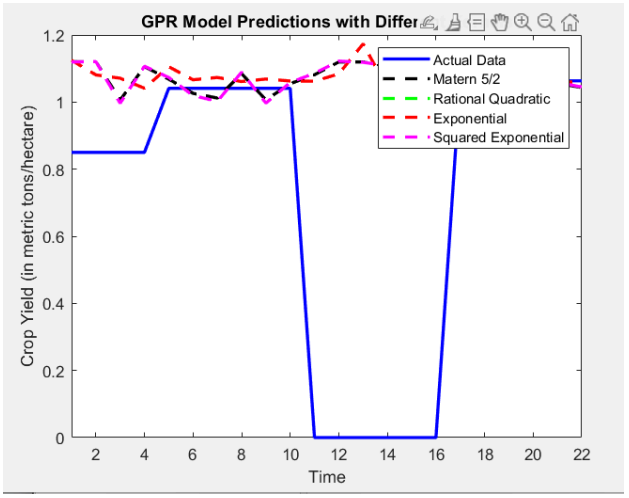
## Batanes



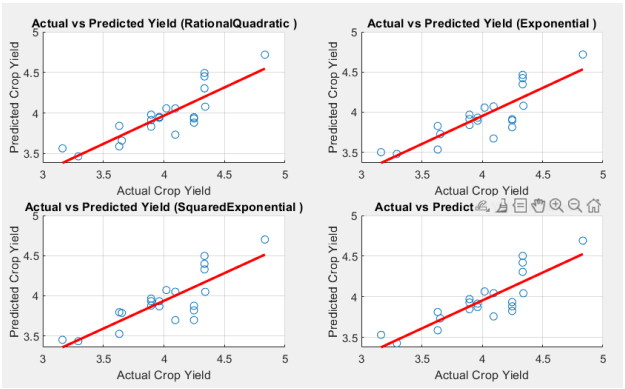

Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | 0.068642 | Inf | 0.088952 | 0.29825 | 0.42038 | 0.74298 |
| {'Exponential' } | 0.067684 | Inf | 0.080635 | 0.28396 | 0.47457 | 0.70739 |
| {'SquaredExponential'} | 0.068642 | Inf | 0.088952 | 0.29825 | 0.42038 | 0.74298 |
| {'Matern52' } | 0.069531 | Inf | 0.085076 | 0.29168 | 0.44563 | 0.72661 |



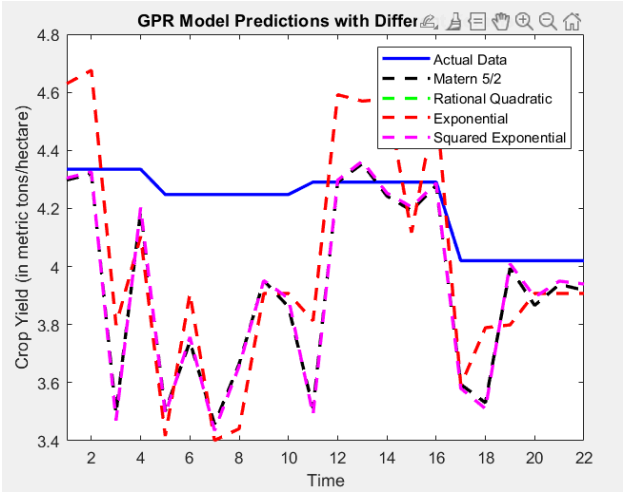




## Cagayan

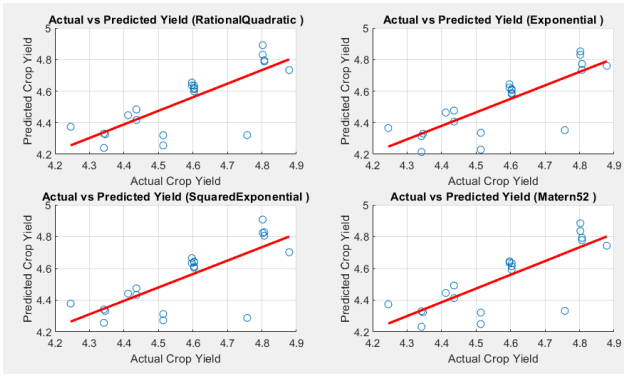



Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | -0.034973 | 3.7836 | 0.039511 | 0.19877 | 0.72147 | 0.51504 |
| {'Exponential' } | -0.050149 | 4.0283 | 0.044278 | 0.21042 | 0.68786 | 0.54523 |
| {'SquaredExponential'} | -0.064393 | 4.2629 | 0.052224 | 0.22853 | 0.63184 | 0.59214 |
| {'Matern52' } | -0.047772 | 4.0192 | 0.042247 | 0.20554 | 0.70218 | 0.53258 |

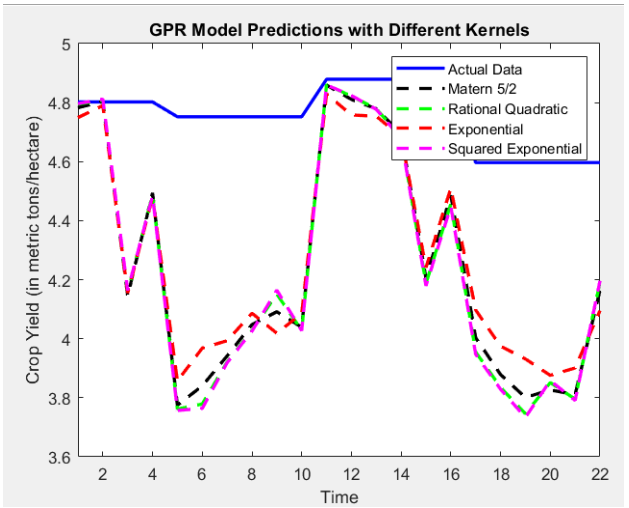



## Isabela

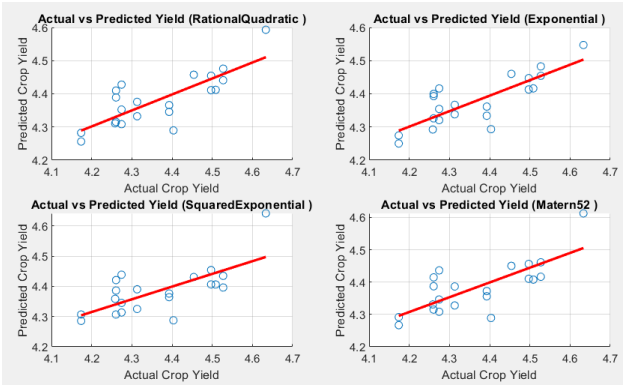



Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | -0.035144 | 1.7615 | 0.017227 | 0.13125 | 0.47056 | 0.71009 |
| {'Exponential' } | -0.045323 | 1.7711 | 0.016245 | 0.12745 | 0.50075 | 0.68955 |
| {'SquaredExponential'} | -0.031216 | 1.788 | 0.018972 | 0.13774 | 0.41694 | 0.74518 |
| {'Matern52' } | -0.036688 | 1.7753 | 0.016849 | 0.1298 | 0.48219 | 0.70225 |

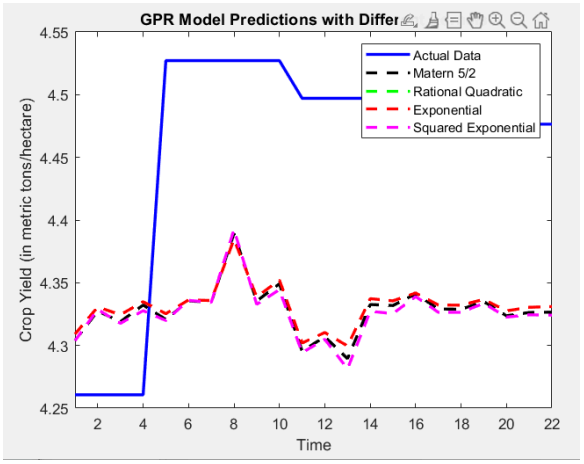


## Nueva Vizcaya

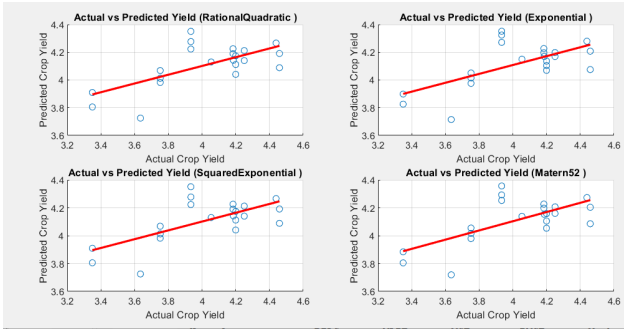



Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | 0.016007 | 1.6653 | 0.0068694 | 0.082882 | 0.57345 | 0.63737 |
| {'Exponential' } | 0.012889 | 1.6762 | 0.0066543 | 0.081574 | 0.58681 | 0.62731 |
| {'SquaredExponential'} | 0.019148 | 1.8587 | 0.008798 | 0.093797 | 0.45369 | 0.72131 |
| {'Matern52' } | 0.01766 | 1.7304 | 0.0076128 | 0.087252 | 0.52728 | 0.67097 |

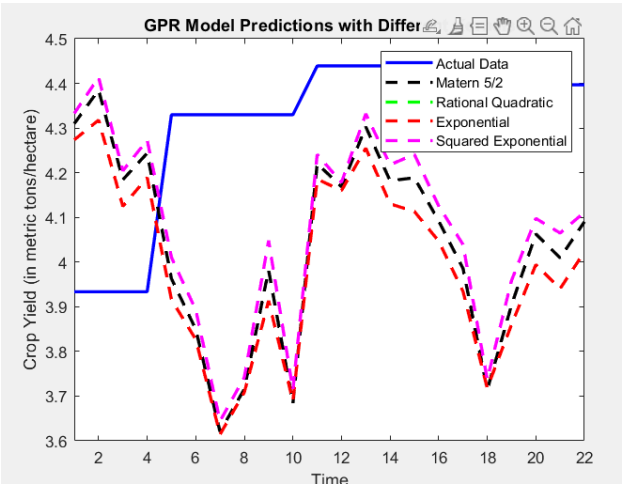



## Quirino

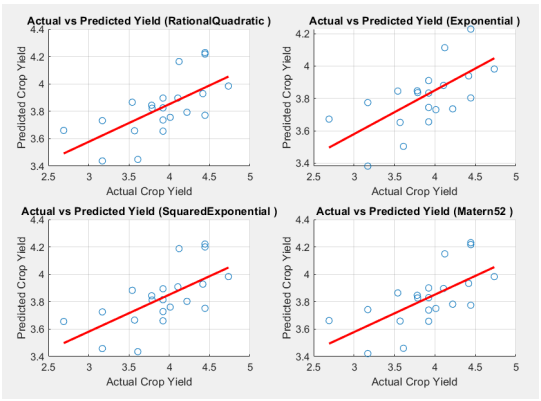



Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | 0.086334 | 5.3939 | 0.067719 | 0.26023 | 0.33296 | 0.79704 |
| {'Exponential' } | 0.09324 | 5.4688 | 0.070092 | 0.26475 | 0.30958 | 0.81089 |
| {'SquaredExponential'} | 0.086334 | 5.3939 | 0.067719 | 0.26023 | 0.33296 | 0.79704 |
| {'Matern52' } | 0.08895 | 5.3804 | 0.067034 | 0.25891 | 0.3397 | 0.793 |



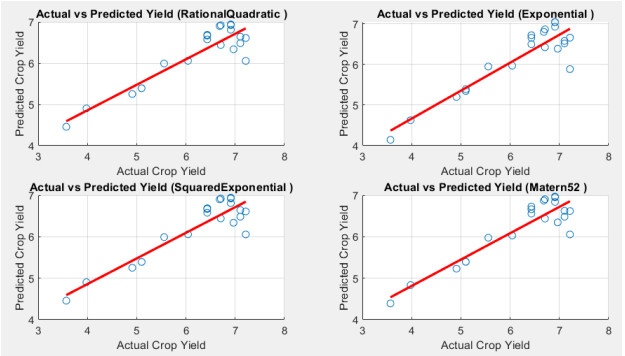

## Aurora




Evaluation Metrics for GPR Models:

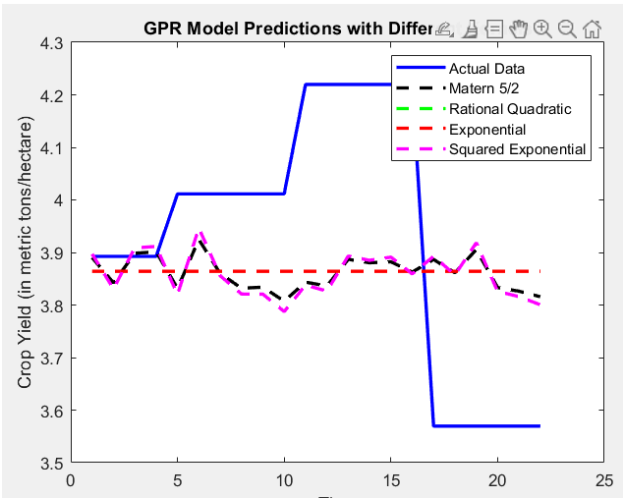
| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | -0.077278 | 8.1045 | 0.15387 | 0.39227 | 0.3453 | 0.78964 |
| {'Exponential' } | -0.07862 | 8.0039 | 0.1555 | 0.39434 | 0.33836 | 0.79381 |
| {'SquaredExponential'} | -0.077302 | 8.2211 | 0.15587 | 0.39481 | 0.33679 | 0.79475 |
| {'Matern52' } | -0.077013 | 8.0701 | 0.15407 | 0.39252 | 0.34445 | 0.79015 |

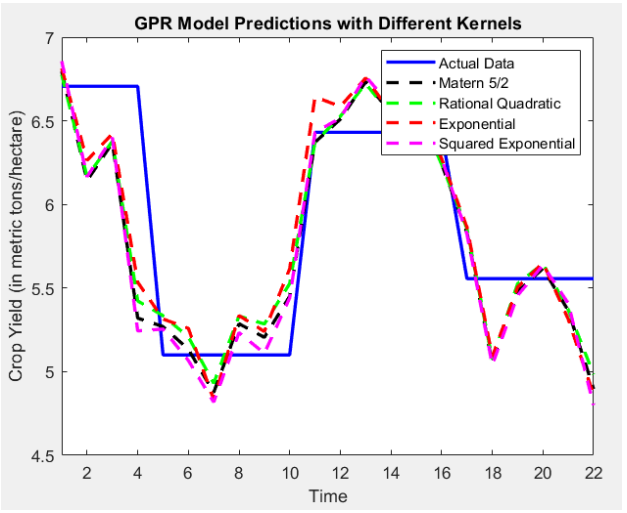
## Nueva Ecija



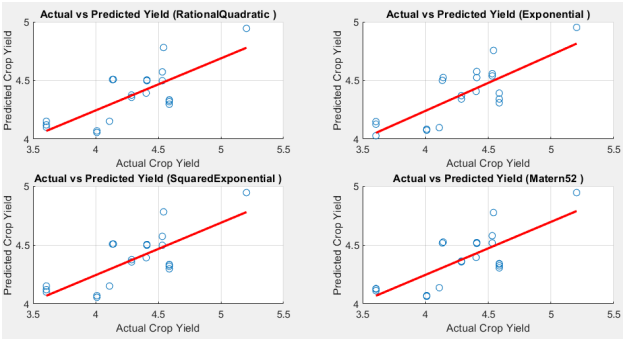

Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | 0.022537 | 6.9377 | 0.24436 | 0.49432 | 0.77752 | 0.46031 |
| {'Exponential' } | -0.017219 | 6.1142 | 0.21216 | 0.46061 | 0.80683 | 0.42892 |
| {'SquaredExponential'} | 0.022547 | 6.9381 | 0.24438 | 0.49435 | 0.77749 | 0.46034 |
| {'Matern52' } | 0.015499 | 6.7342 | 0.23163 | 0.48128 | 0.7891 | 0.44817 |



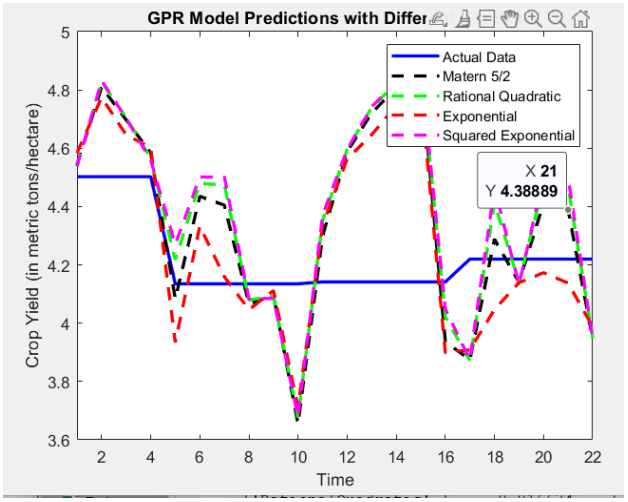




## Tarlac

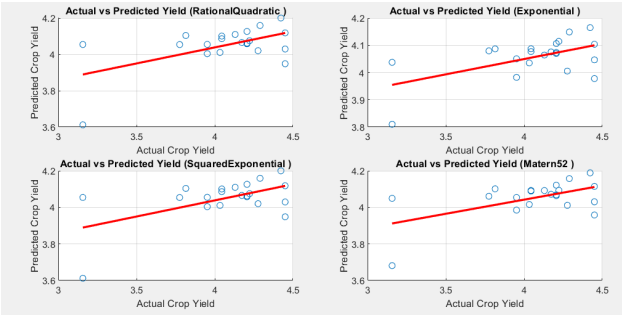



Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | 0.08033 | 5.1435 | 0.072992 | 0.27017 | 0.49745 | 0.69182 |
| {'Exponential' } | 0.086242 | 4.9581 | 0.068202 | 0.26116 | 0.53043 | 0.66874 |
| {'SquaredExponential'} | 0.08033 | 5.1435 | 0.072992 | 0.27017 | 0.49745 | 0.69182 |
| {'Matern52' } | 0.084201 | 5.1261 | 0.072883 | 0.26997 | 0.4982 | 0.69131 |

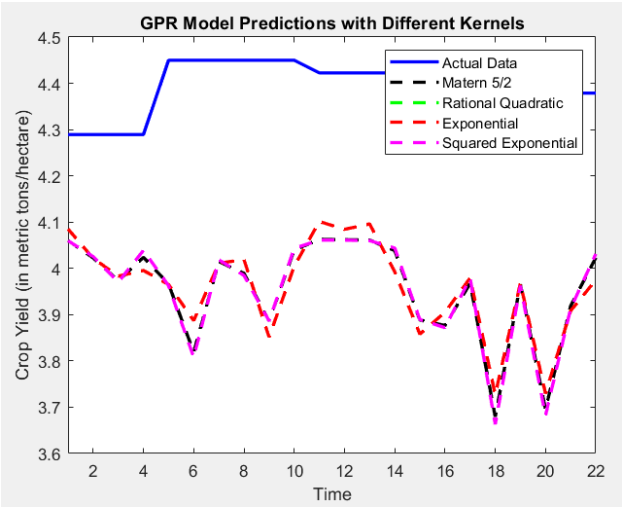



## Zambales



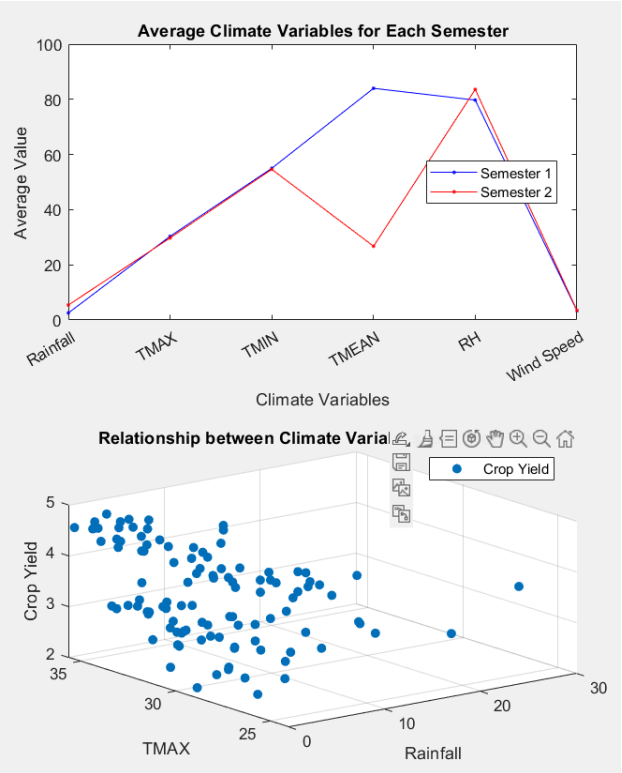

Evaluation Metrics for GPR Models:

| Kernel | BIAS | MAPE | MSE | RMSE | Nash | RRMSE |
|---|---|---|---|---|---|---|
| {'RationalQuadratic' } | -0.016674 | 5.8216 | 0.092594 | 0.30429 | 0.24929 | 0.84555 |
| {'Exponential' } | -0.010192 | 6.0502 | 0.10088 | 0.31762 | 0.18208 | 0.8826 |
| {'SquaredExponential'} | -0.016674 | 5.8216 | 0.092594 | 0.30429 | 0.24929 | 0.84555 |
| {'Matern52' } | -0.014708 | 5.8909 | 0.095141 | 0.30845 | 0.22864 | 0.85711 |



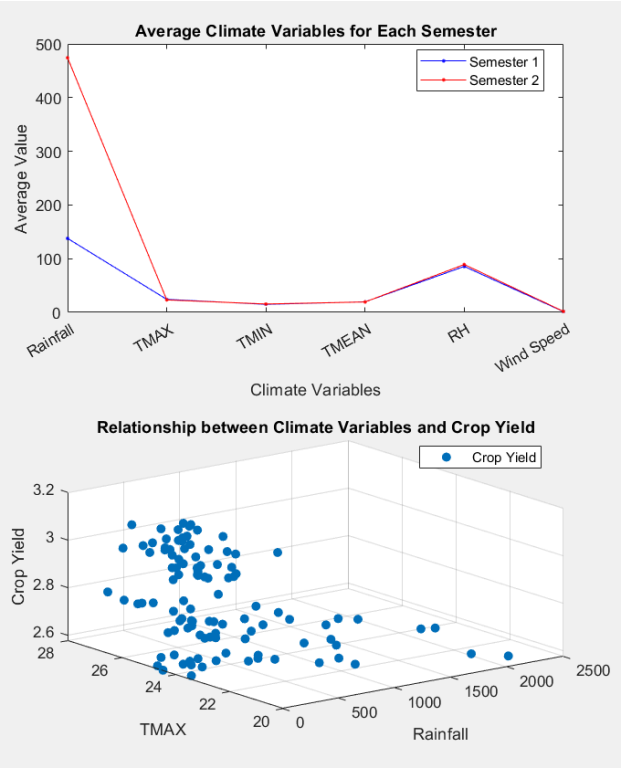
## 6.2. Predicted Optimal Planting Window for each Province

### Apayao



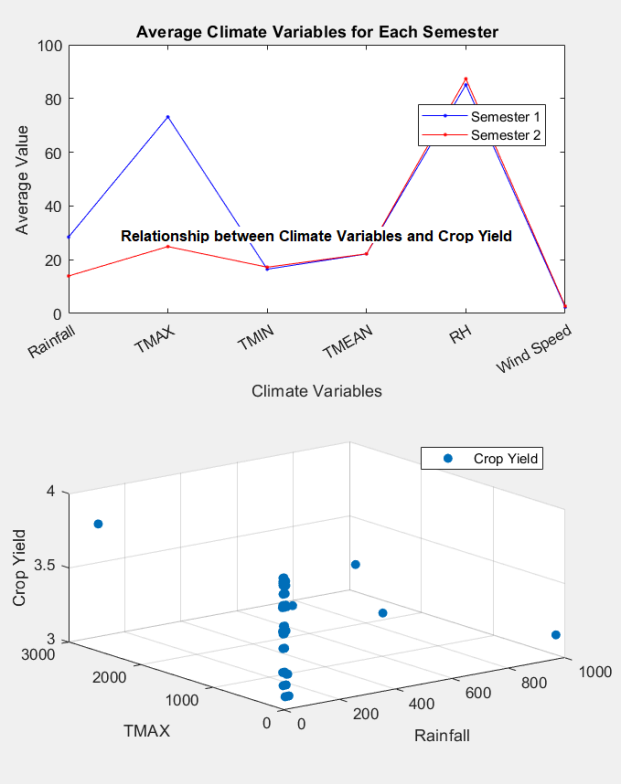

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

### Benguet



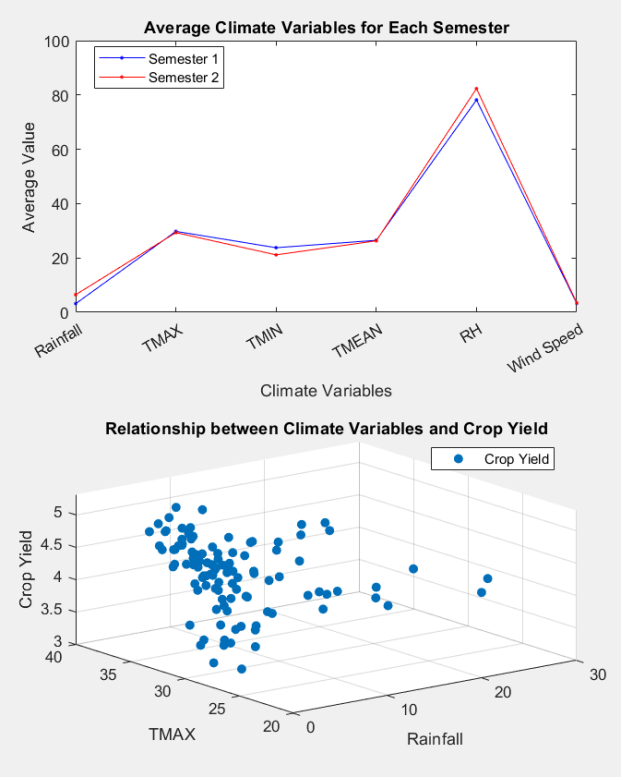

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

### Ifugao



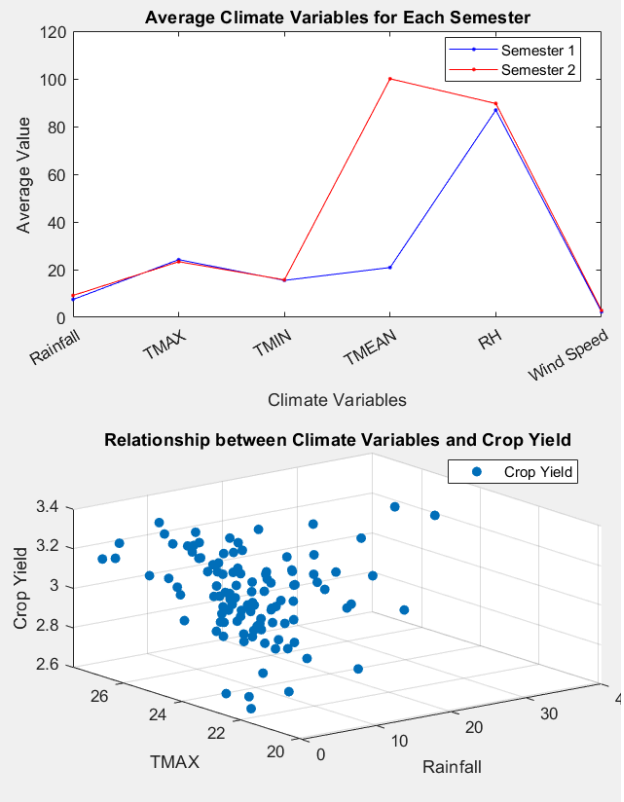

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

### Kalinga



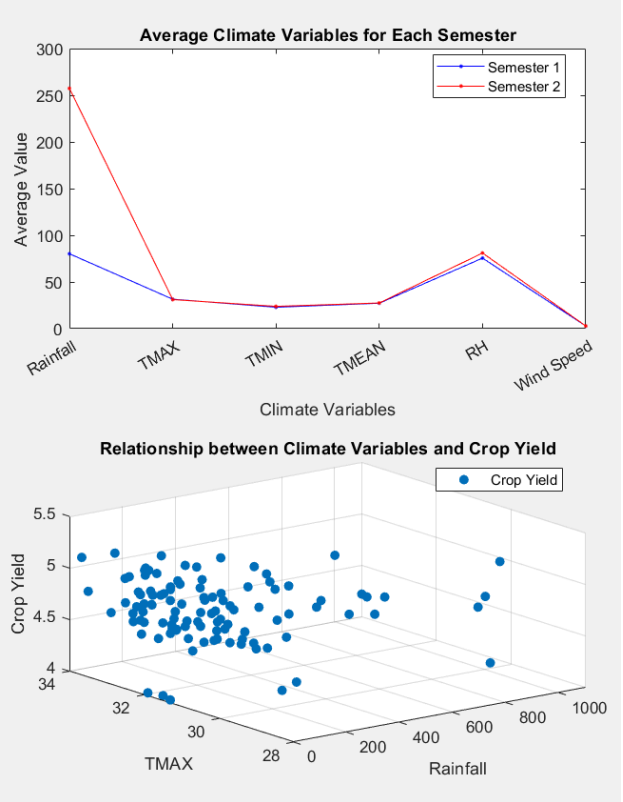

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Mountain Province



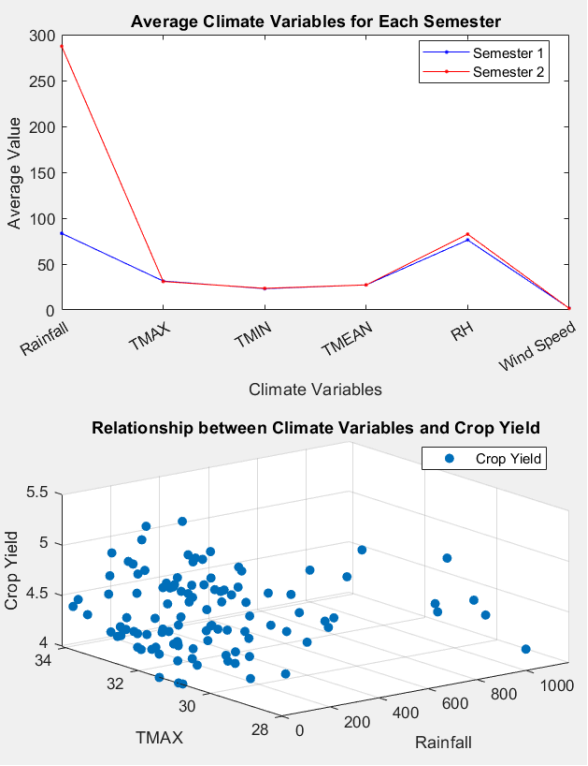

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Ilocos Norte



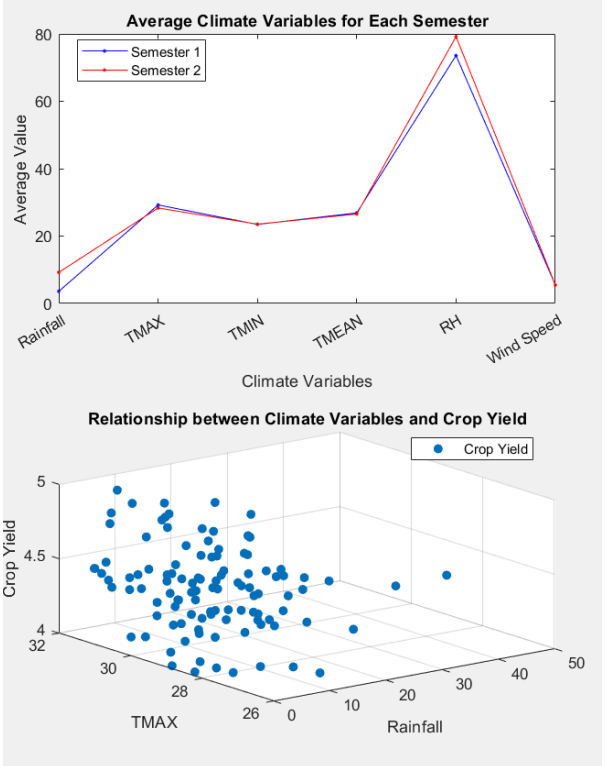

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Ilocos Sur



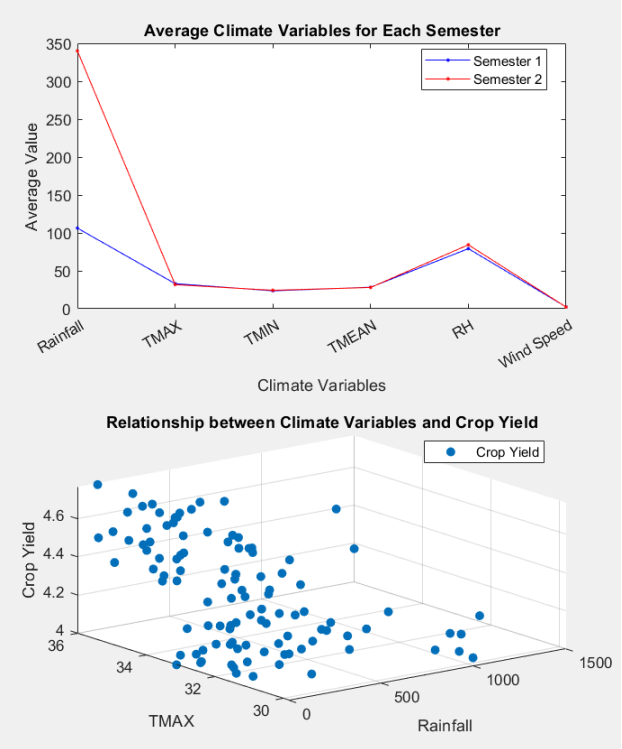

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## La Union



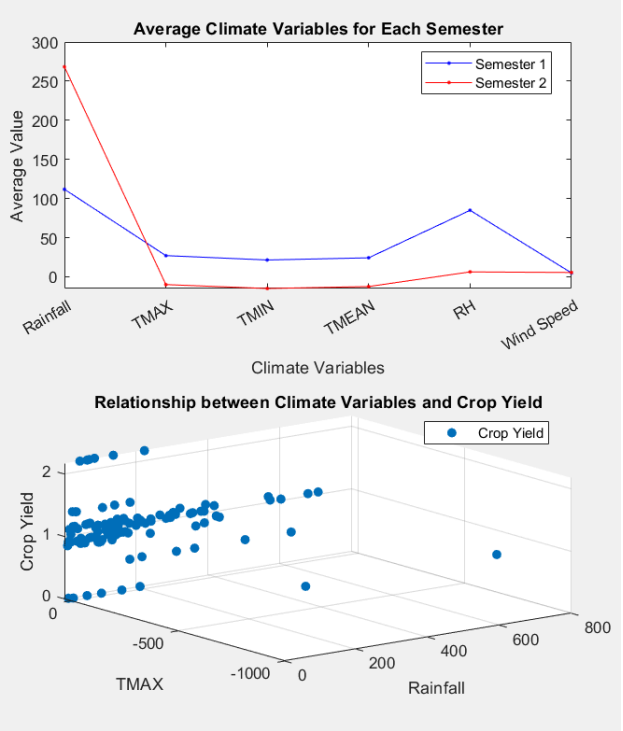

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Pangasinan



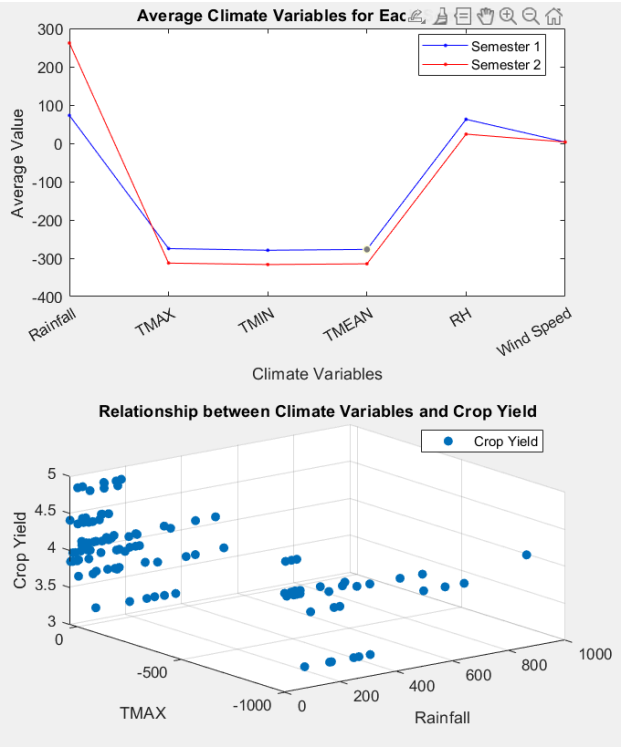

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Batanes



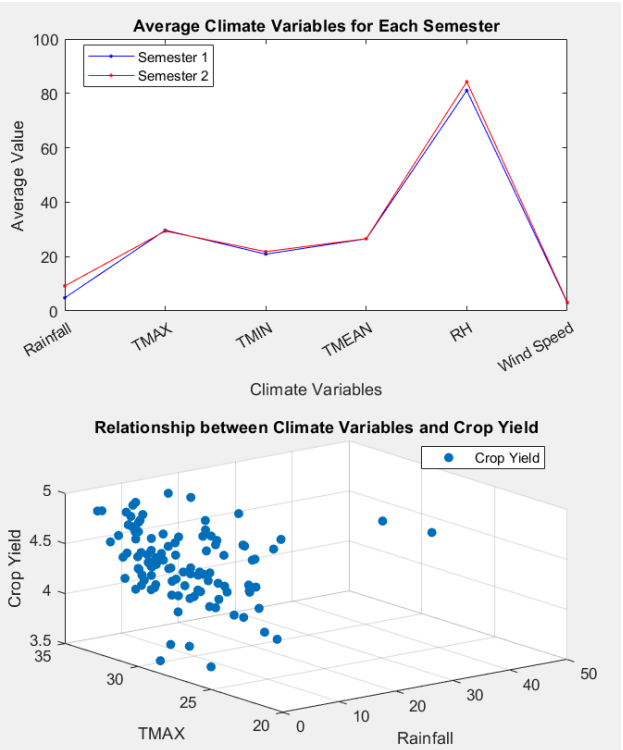

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Cagayan



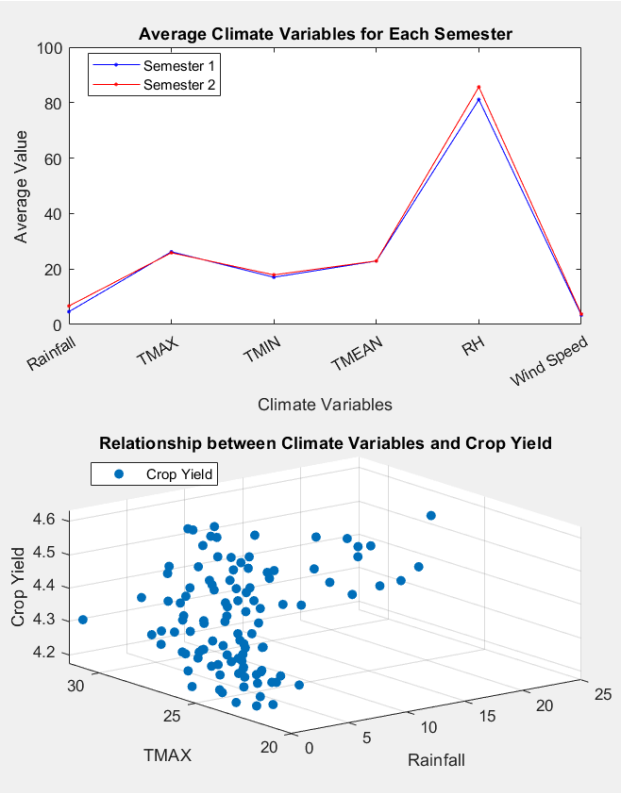

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Isabela



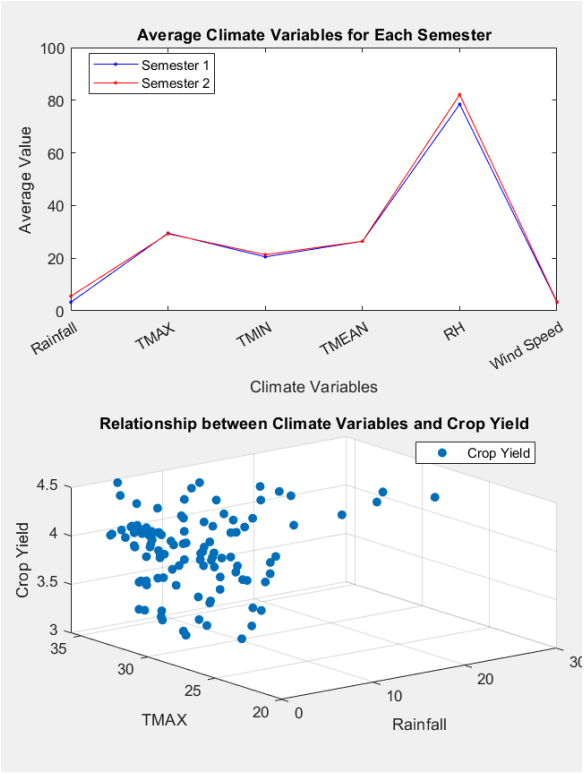

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Nueva Vizcaya



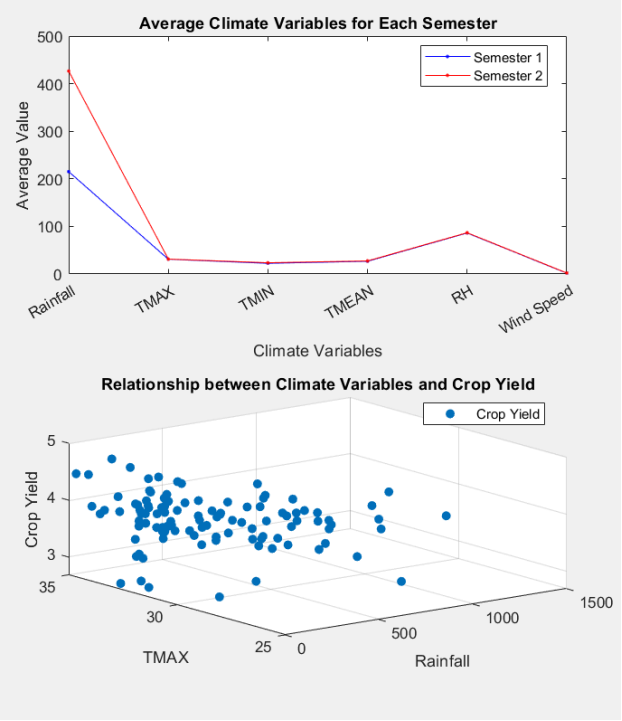

```
>> Prediction
The Optimal Planting Semester is Semester 2
```

## Quirino



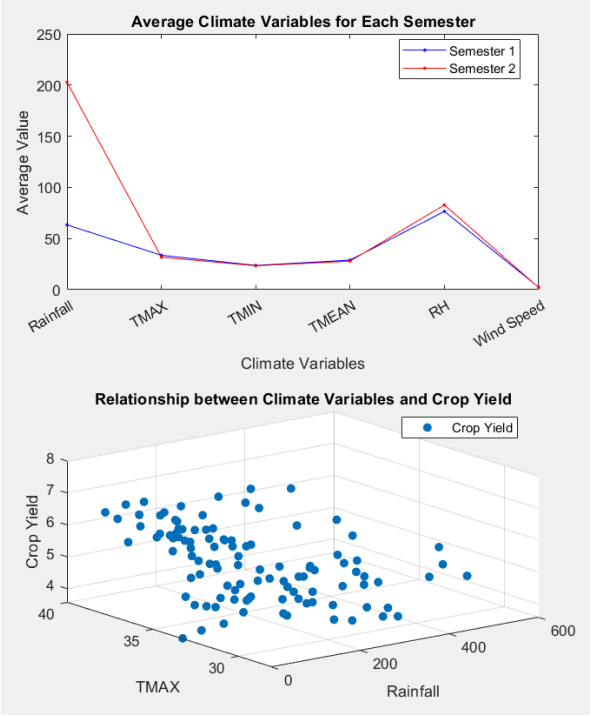

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Aurora



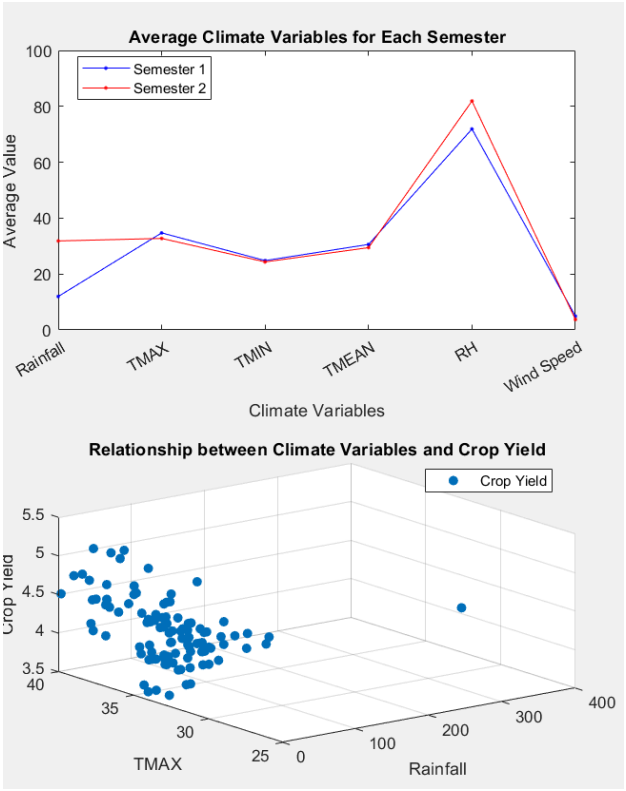

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## Nueva Ecija



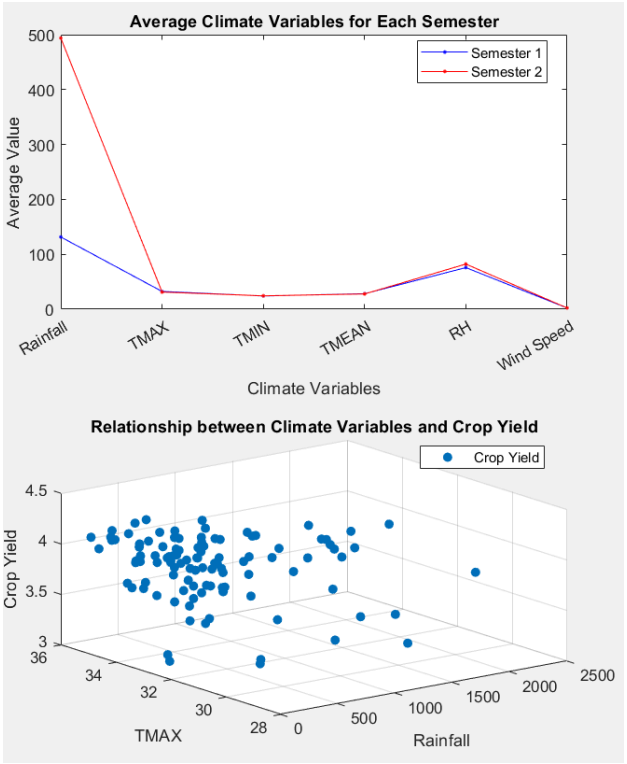

```
>> Prediction
The Optimal Planting Semester is Semester 1
```

**Tarlac**




```
>> Prediction
The Optimal Planting Semester is Semester 1
```

**Zambales**




```
>> Prediction
The Optimal Planting Semester is Semester 1
```

## 6.3. MATLAB Codes

### 6.3.1 Collinearity Code correlation.m

```
data = readtable('Abra.csv'); %Load data

% Climate variables and crop yield
climate_variables = data(:, 3:end-1);
crop_yield = data(:, end);

% Pearson's correlation coefficient
correlation_matrix = corrcoef([table2array(climate_variables),
table2array(crop_yield)]);

% Display the correlation matrix
figure;
h = imagesc(correlation_matrix);
colorbar;
xticks(1:size(correlation_matrix, 2));
xticklabels({'RA', 'Tmax', 'Tmin', 'Tmean', 'RH', 'SPD', 'YLD'});
yticks(1:size(correlation_matrix, 2));
yticklabels({'RA', 'Tmax', 'Tmin', 'Tmean', 'RH', 'SPD', 'YLD'});
title('Collinearity Matrix of Climate Variables and Crop Yield');
textStrings = num2str(correlation_matrix(:), '%0.2f');
textStrings = strtrim(cellstr(textStrings));
[x, y] = meshgrid(1:size(correlation_matrix, 2));
hText = text(x(:), y(:), textStrings, 'HorizontalAlignment', 'center',
'FontSize', 8);
```

### 6.3.2 GPR Program GPRmod.m (includes evaluation metrics)

```
% Read data from CSV file
data = readmatrix('Abra.csv');

X = data(:, 1:end-1); % Climate variables
Y = data(:, end); % Crop yield
X_normalized = normalize(X); % Normalize features

% Train-Test Split (80% train, 20% test)
cv = cvpartition(size(X, 1), 'Holdout', 0.2);
X_train = X_normalized(cv.training, :);
Y_train = Y(cv.training);
X_test = X_normalized(cv.test, :);
Y_test = Y(cv.test);

% Gaussian Process Regression model
kernelNames = {'RationalQuadratic', 'Exponential', 'SquaredExponential',
'Matern52'};

% Array for Evaluation Metrics
bias = zeros(1, length(kernelNames));
mape = zeros(1, length(kernelNames));
mse = zeros(1, length(kernelNames));
rmse = zeros(1, length(kernelNames));
nash = zeros(1, length(kernelNames));
rrmse = zeros(1, length(kernelNames));

% Table
figure;
for i = 1:length(kernelNames)

   % Gaussian Process Regression model
   model = fitrgp(X_train, Y_train, 'KernelFunction', kernelNames{i});

   % Prediction on test set
   YPred = predict(model, X_test);

   % Evaluation metrics Calculation
   bias(i) = mean(YPred - Y_test);
   mape(i) = mean(abs((YPred - Y_test) ./ Y_test)) * 100;
   mse(i) = mean((YPred - Y_test).^2);
   rmse(i) = sqrt(mse(i));
   nash(i) = 1 - sum((YPred - Y_test).^2) / sum((mean(Y_test) - Y_test).^2);
   rrmse(i) = rmse(i) / std(Y_test);

   % Linear regression line parameters
   p = polyfit(Y_test, YPred, 1);
   Y_reg = polyval(p, Y_test);

   % Plot actual vs predicted crop yield with regression line
   subplot(2, 2, i);
   scatter(Y_test, YPred);
   hold on;
   plot(Y_test, Y_reg, 'r-', 'LineWidth', 2);
   hold off;
   xlabel('Actual Crop Yield');
   ylabel('Predicted Crop Yield');
   title(['Actual vs Predicted Yield (', kernelNames{i}, ' )']);
   grid on;
end

evaluationTable = table(kernelNames', bias', mape', mse', rmse', nash', rrmse',
...
```

```
    'VariableNames', {'Kernel', 'BIAS', 'MAPE', 'MSE', 'RMSE', 'Nash', 'RRMSE'});

disp('Evaluation Metrics for GPR Models:');
disp(evaluationTable);

% Trend Line Graph

climate_variable_columns = [2, 3, 4];
yield_column = 1;
your_climate_data = data(:, 1:end-1);
your_yield_data = data(:, end);

% Split Data
train_fraction = 0.8;
train_size = round(train_fraction * size(data, 1));
train_data = your_yield_data(1:train_size, :);
test_data = your_yield_data(train_size+1:end, :);
train_features = your_climate_data(1:train_size, :);
test_features = your_climate_data(train_size+1:end, :);

% Train GPR Model with Matern 5/2 Kernel
gpr_model_matern = fitrgp(train_features, train_data, 'KernelFunction',
'matern52');

% Train GPR Model with Rational Quadratic Kernel
gpr_model_rational = fitrgp(train_features, train_data, 'KernelFunction',
'rationalquadratic');

% Train GPR Model with Exponential Kernel
gpr_model_exponential = fitrgp(train_features, train_data, 'KernelFunction',
'exponential');

% Train GPR Model with Squared Exponential Kernel
gpr_model_squared_exponential = fitrgp(train_features, train_data,
'KernelFunction', 'squaredexponential');

% Predictions
predictions_matern = predict(gpr_model_matern, test_features);
predictions_rational = predict(gpr_model_rational, test_features);
predictions_exponential = predict(gpr_model_exponential, test_features);
predictions_squared_exponential = predict(gpr_model_squared_exponential,
test_features);

% Graph Results
figure;
plot(1:numel(test_data), test_data{:,:}, 'b', 'LineWidth', 2);
hold on;
plot(1:numel(predictions_matern), predictions_matern, 'k--', 'LineWidth', 2);
plot(1:numel(predictions_rational), predictions_rational, 'g--', 'LineWidth', 2);
plot(1:numel(predictions_exponential), predictions_exponential, 'r--',
'LineWidth', 2);
plot(1:numel(predictions_squared_exponential), predictions_squared_exponential,
'm--', 'LineWidth', 2);
xlabel('Time');
ylabel('Crop Yield (in metric tons/hectare)');
legend('Actual Data', 'Matern 5/2', 'Rational Quadratic', 'Exponential', 'Squared
Exponential');
title('GPR Model Predictions with Different Kernels');
xlim([start_time_index, end_time_index]);
hold off;
```

**6.3.3 Prediction Program** Prediction.m

```
data = readtable('Abra.csv');

% Columns
year = data.YEAR;
month = data.MONTH;
rainfall = data.RAINFALL;
tmax = data.TMAX;
tmin = data.TMIN;
tmean = data.TMEAN;
rh = data.RH;
wind_speed = data.SPD;
ry = data.YLD;

% Climate Matric
climate_data = [rainfall, tmax, tmin, tmean, rh, wind_speed];

% Average climate variables for each semester
semester1_idx = (month >= 1 & month <= 6);
semester2_idx = (month >= 7 & month <= 12);
avg_climate_semester1 = mean(climate_data(semester1_idx, :));
avg_climate_semester2 = mean(climate_data(semester2_idx, :));

% Average crop yield for each semester
avg_yield_semester1 = mean(ry(semester1_idx));
avg_yield_semester2 = mean(ry(semester2_idx));

% Plot: Relationship between climate variables and crop yield
figure;
subplot(2, 1, 1);
climate_variables = {'Rainfall', 'TMAX', 'TMIN', 'TMEAN', 'RH', 'Wind Speed'};
plot(1:6, avg_climate_semester1, 'b.-', 'DisplayName', 'Semester 1');
hold on;
plot(1:6, avg_climate_semester2, 'r.-', 'DisplayName', 'Semester 2');
title('Average Climate Variables for Each Semester');
xlabel('Climate Variables');
ylabel('Average Value');
xticks(1:6);
xticklabels(climate_variables);
legend('Location', 'Best');
subplot(2, 1, 2);
scatter3(climate_data(:, 1), climate_data(:, 2), ry, 'filled', 'DisplayName',
'Crop Yield');
title('Relationship between Climate Variables and Crop Yield');
xlabel('Rainfall');
ylabel('TMAX');
zlabel('Crop Yield');
legend('Location', 'Best');

% Find optimal planting semester based on maximum average yield
if avg_yield_semester1 > avg_yield_semester2
   optimal_semester = 'Semester 1';
else
   optimal_semester = 'Semester 2';
end
fprintf('The Optimal Planting Semester is %s\n', optimal_semester);
```